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PNC ULTRA-LMR MASTER COMPENDIUM • NUR-10 • VERIFIED

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Batch 1	I ENT, II Eye, III Renal, IV Male repro	18
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MASTER TABLE OF CONTENTS — NUR-10 AHN-II

Pg	Unit	Focus
4–21	I–IV (B1)	ENT/trach/hearing, eye/cataract/glaucoma, AKI/CKD/dialysis/electrolytes, male repro
22–39	V–VI-A (B2)	Burns/Parkland/grafts/NPWT, GCS/ICP/CPP/SCI/stroke/seizure/SAH/meningitis
40–57	VI-B–VIII (B3)	Parkinson/Alzheimer/GBS/MG/MS/rehab, HIV/ART/NACO/anaphylaxis, cancer/staging/chemo/radio/BMT/emergencies/palliative
58–75	IX–XII (B4)	ABCDE/polytrauma/bites/poison/thermal/disaster/triage/MLC, aging/CGA/abuse, ICU/vent/alarms/hemodynamics/defib/sepsis/ACLS, EOL/occupational

How to use: 1 concept/page • quick-strip first • trap vs hit boxes • speed-check before footer • urine output guides burns, CPP guides neuro, cultures<1h guide sepsis, swallow screen guides stroke. Double-QA pending sign-off on ACLS/stroke/dialysis/burns/HIV/triage/vent per current guidance.

DOUBLE-QA CHECKLIST (Verifier 1 / Verifier 2)

Q	Standard	Status
Q1	12 units, 140T+40SL+480 clinical mapped	Done — batches 1-4
Q2	AHN-I overlap avoided	Done
Q3	Renal/dialysis/electrolytes current	Needs sign-off
Q4	GCS/ICP/CP/Stroke/GBS/MG correct	Needs sign-off
Q5	TBSA/Parkland current	Needs sign-off
Q6	HIV/NACO current	Needs sign-off
Q7	Chemo/RT/BMT/emergencies current	Needs sign-off
Q8	Triage identified/current	Needs sign-off
Q9	Geriatrics programmes current	Done
Q10	Vent/defib/ACLS/sepsis current	Needs sign-off
Q11	Palliative integrated	Done
Q12	One concept/page, large visuals	Done

ENT ASSESSMENT, AUDIOMETRY & WEBER/RINNE TESTS

ENT ASSESSMENT: Otitoscopic exam (pull pinna **up and back** in adults; down and back in infants < 3 yrs). Tuning fork tests (512 Hz): Weber tests lateralization; Rinne compares Air Conduction (AC) to Bone Conduction (BC)!

TUNING FORK TEST	PROCEDURAL PROTOCOL & NORMAL PHYSIOLOGICAL FINDING	ABNORMAL PATHOLOGY & INTERPRETATION
RINNE TEST (AC vs BC)	Place vibrating fork on mastoid bone until sound ceases, then move immediately 1–2 cm in front of ear canal. • Normal (Positive Rinne): Air Conduction exceeds Bone Conduction (AC > BC by 2:1 ratio).	• Conductive Loss (Negative Rinne): BC > AC (Sound heard longer on mastoid). • Sensorineural Loss: AC > BC, but overall duration is reduced.
WEBER TEST (Lateralization)	Place vibrating fork on midline of forehead or top of skull. Ask client where sound is heard louder (left, right, or equally).	• Normal: Heard equally in both ears (no lateralization). • Conductive Loss: Sound lateralizes to the DEFECTIVE (BAD) EAR (masked from ambient noise). • Sensorineural Loss: Sound lateralizes to the NORMAL (GOOD) EAR .

AUDIOMETRY & OTOSCOPY FUNDAMENTALS

OTOSCOPIC EXAMINATION TECHNIQUE

- Adult: Pull pinna **UP and BACK** to straighten external auditory canal.
- Infant / Child (< 3 yrs): Pull pinna **DOWN and BACK**.
- Inspect tympanic membrane (TM): Pearly gray, translucent, intact, with visible light reflex (cone of light at 5 o'clock right, 7 o'clock left).

PURE-TONE AUDIOMETRY

- Measures hearing threshold across multiple sound frequencies (Hz) and decibels (dB).
- Conductive hearing loss impairs low frequencies; Sensorineural loss impairs high frequencies.

OTOSCOPIC PEDIATRIC DIRECTION TRAP

Pulling the pinna *up and back* on an infant during otoscopy fails to straighten the immature ear canal, causing poor visualization and potential canal trauma. Always pull **down and back** in children under 3!

 ENT Assessment Speed Check

What is the normal physiologic ratio of Air Conduction to Bone Conduction on the Rinne test?

Air Conduction is greater than Bone Conduction (AC > BC by 2:1 ratio)

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NUR-09 Batch 1: Units I, II, III & IV (Sensory & Renal)

IMPACTED WAX, TM PERFORATION & OTITIS MEDIA

⚡ **EAR DISORDERS:** Impacted cerumen requires warm irrigation (contraindicated if TM perforated!); Otitis Media (Acute vs Chronic) causes severe otalgia and purulent otorrhea. **Mastoiditis is a dangerous complication of untreated OM!**

CONDITION	PATHOLOGICAL ETIOLOGY & MECHANISM	CLINICAL PRESENTATION	NURSING PRIORITY
IMPACTED CERUMEN	Excessive earwax accumulation or improper Q-tip usage pushing wax against tympanic membrane.	Conductive hearing loss, tinnitus, otalgia, fullness sensation in ear.	Warm saline irrigation (body temp); gentle cerumenolytic drops (mineral oil/carbamide peroxide).
ACUTE OTITIS MEDIA (AOM)	Bacterial/viral infection of middle ear space behind tympanic membrane (often following upper respiratory infection).	Throbbing severe otalgia, fever, bulging erythematous TM, diminished light reflex, irritability in infants.	Analgesics, systemic antibiotics (Amoxicillin), warm compresses.
CHRONIC SUPPURATIVE OM	Chronic inflammation with persistent TM perforation and purulent foul-smelling otorrhea (> 6 weeks).	Persistent conductive hearing loss, intermittent purulent ear discharge, tympanic membrane defect.	Ear wicks with antibiotic drops, Tympanoplasty surgical repair.

MYRINGOTOMY & TYMPANOSTOMY TUBES

MYRINGOTOMY PROCEDURE

- Surgical incision of tympanic membrane to relieve pressure and drain purulent fluid in severe OM.
- Insertion of tiny pressure-equalizing (PE) grommet tubes.

POST-OP EAR PROTECTION

- Keep ears dry during bathing/swimming (use silicone earplugs or shower caps).
- Report purulent drainage or sudden hearing loss immediately.

IRRIGATION WITH PERFORATED TM TRAP

Irrigating the external ear canal when the tympanic membrane is perforated or ruptured forces water, bacteria, and debris straight into the middle ear, triggering acute mastoiditis and severe sepsis! Always inspect TM via otoscope before irrigating.

 **Ear Disorders Speed Check**

What is the surgical incision of the tympanic membrane to relieve middle ear pressure termed?

Myringotomy (With or without tympanostomy tube insertion)

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MENIERE'S DISEASE, LABYRINTHITIS & OTOTOXICITY

 **MENIERE'S DISEASE:** Inner ear disorder characterized by endolymphatic hydrops (excess fluid in cochlea/labyrinth). **Classic Triad: Episodic Vertigo, Tinnitus, and Fluctuating Sensorineural Hearing Loss!** Fall safety is paramount.

INNER EAR DISORDER	PATHOPHYSIOLOGY & DIAGNOSTIC HALLMARKS	CLINICAL NURSING MANAGEMENT
MENIERE'S DISEASE	Distension of endolymphatic system in inner ear. Sudden debilitating vertigo attacks lasting hours, accompanied by nausea, vomiting, nystagmus, and roaring tinnitus.	• Strict Low-Sodium Diet (< 1,500–2,000 mg/day) to reduce endolymphatic fluid pressure. • Diuretics (Hydrochlorothiazide), Antiemetics, Meclizine. • Fall precautions during acute vertigo attacks.
LABYRINTHITIS	Inflammation of the inner ear labyrinth (viral or bacterial infection). Sudden severe vertigo, nausea, and hearing loss.	Antihistamines, corticosteroids, vestibular rehabilitation, hydration, and safety precautions.
OTOTOXICITY	Drug-induced damage to the 8th Cranial Nerve (Vestibulocochlear) or inner ear structures, causing permanent tinnitus and hearing loss.	• Culprit Drugs: Aminoglycosides (Gentamicin, Tobramycin), Loop Diuretics (Furosemide IV push), Vancomycin, Cisplatin, Aspirin. • Monitor baseline/serial audiograms and peak/trough levels.

FALL PREVENTION & VERTIGO ATTACK MANAGEMENT

ACUTE VERTIGO PROTOCOLS

- Instruct client to lie flat in a dark, quiet room during acute vertigo attacks.
- Avoid sudden head turns, bright flashing lights, or rapid postural changes.
- Keep emesis basin nearby.

OTOTOXIC DRUG MONITORING

- Tinnitus or high-frequency hearing loss is often the **very first sign of impending ototoxicity!**
- Stop drug immediately and report to provider at first complaint of ringing in ears.



Inner Ear Speed Check

What dietary modification is prescribed for patients diagnosed with Meniere's disease?

Strict Low-Sodium Diet (Reduces endolymphatic fluid pressure and hydrops)

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RHINITIS, TONSILLITIS & POST-TONSILLECTOMY CARE

 **UPPER AIRWAY & TONSILLECTOMY:** Pharyngitis and tonsillitis require monitoring for airway compromise. **Post-Tonsillectomy Hemorrhage (PTH)** can occur suddenly on Post-Op Day 5 to 7 when the surgical slough/eschar falls off!

CONDITION / PROCEDURE	CLINICAL FEATURES & SURGICAL PROTOCOL	CRITICAL NURSING INTERVENTION
ACUTE TONSILLITIS	Inflammation of palatine tonsils (viral or Group A beta-hemolytic Streptococcus). Sore throat, dysphagia, exudate, fever, cervical lymphadenopathy.	Penicillin antibiotic course (prevent rheumatic fever/glomerulonephritis), warm saline gargles, analgesics.
POST-TONSILLECTOMY CARE	Surgical removal of palatine tonsils. Immediate post-op priority: Maintain patent airway ; position on side/prone until fully awake.	• Frequent swallowing is the #1 sign of bleeding! • Ice collar to neck; clear liquids/ice cream (avoid red/brown fluids!).

POST-TONSILLECTOMY HEMORRHAGE (PTH) EMERGENCY

EARLY WARNING SIGNS OF BLEEDING

- • **Frequent swallowing or throat clearing** while sleeping or awake.
- • Bright red blood trickling from mouth or emesis of dark blood (coffee ground / frank red).
- • Tachycardia, restlessness, pallor.

EMERGENCY ACTION PLAN

- 1. Position client in side-lying or semi-Fowler's position.
- 2. Notify surgeon/ENT specialist immediately.
- 3. Keep suction apparatus ready at bedside. Prepare for emergency return to OR for suture ligation!

POST-OP RED FLUID TRAP

Offering red-colored juices, gelatin, or popsicles to a pediatric patient post-tonsillectomy is a **clinical trap!** Red food coloring masks fresh blood in emesis, obscuring the detection of active post-operative hemorrhage.



Upper Airway Speed Check

What is the **earliest subtle clinical sign** of post-tonsillectomy hemorrhage?

Frequent swallowing or constant throat clearing while resting

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EPISTAXIS PATHOPHYSIOLOGY & EMERGENCY MANAGEMENT

 **EPISTAXIS (NOSEBLEED):** Rupture of superficial mucosal capillaries. **#1 Most common site: Kiesselbach's Plexus (Little's Area) on anterior nasal septum.** Posterior nosebleeds are severe medical emergencies requiring posterior nasal packing!

EPISTAXIS TYPE	ANATOMICAL SOURCE & CHARACTERISTICS	EMERGENCY NURSING MANAGEMENT
ANTERIOR EPISTAXIS (90% of cases)	Kiesselbach's Plexus (vascular network on anterior nasal septum). Easy to visualize and control.	• Sit upright and lean forward slightly (prevents swallowing blood into stomach!). • Pinch soft lower portion of nose continuously for 10–15 mins. • Apply ice pack to bridge of nose.
POSTERIOR EPISTAXIS (Severe / Elderly)	Sphenopalatine artery branches deep in nasal cavity. Bleeding flows backward into pharynx; massive risk of aspiration and shock.	• Requires posterior nasal packing (balloon catheter / Foley catheter tamponade). • Continuous hemodynamic monitoring and supplemental oxygen. • Monitor for respiratory depression.

POST-PACKING NURSING CARE & COMPLICATIONS

POSTERIOR NASAL PACKING RISKS

- Posterior nasal packs obstruct upper airway and impair gas exchange.
- **Complication:** Hypoxia, hypoventilation, toxic shock syndrome.
- Monitor SaO₂ continuously; administer humidified O₂; keep wire cutters/scissors at bedside.

PATIENT INSTRUCTIONS

- Instruct client to breathe through the mouth.
- Avoid nose blowing, straining, or vigorous coughing for 48–72 hours post-packing.
- Avoid hot liquids, spicy foods, and aspirin/NSAIDs.

EPISTAXIS HEAD TILT BACK TRAP

Tilting a patient's head *backward* during a nosebleed causes blood to flow down the posterior pharynx into the stomach, leading to **nausea, vomiting, aspiration, and inability to estimate blood loss!** Always lean forward.



Epistaxis Speed Check

What is the correct patient positioning during an acute anterior nosebleed?

Sitting upright and leaning forward slightly with direct nasal bridge compression

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NASAL/LARYNGEAL OBSTRUCTION & TRACHEOSTOMY CARE

 **TRACHEOSTOMY CARE:** Surgical airway providing direct tracheal access. **Crucial bedside safety gear: Obturator + Spare tracheostomy tube (same size + one size smaller) + Tracheal dilator!** Cuff pressure maintained at 20–25 mmHg.

TRACH COMPONENT	FUNCTIONAL MECHANISM & CARE PROTOCOL	CRITICAL NURSING PRECAUTION
OBTURATOR	Guide inserted inside outer cannula during insertion to prevent mucosal trauma.	REMOVE IMMEDIATELY AFTER INSERTION! Tape to head of bed for emergency re-insertion.
CUFF MANAGEMENT	Inflatable balloon seals trachea for mechanical ventilation and prevents aspiration.	Maintain cuff pressure at 20 to 25 mmHg (25–30 cmH2O) . Over-inflation causes tracheal necrosis and fistula.
INNER CANNULA	Removable sleeve that catches secretions. Can be cleaned or replaced.	Clean with sterile hydrogen peroxide and saline; inspect q8h for mucus plugging.

TRACHEOSTOMY EMERGENCY PROTOCOLS

ACCIDENTAL DECANNULATION (< 72 HRS)

- A fresh tracheostomy tract is unhealed and prone to collapsing.
- ✓ **Action:** Call for emergency medical help / surgeon immediately! **DO NOT attempt blind re-insertion** (creates false passage). Use tracheal dilator.

SUCTIONING SAFETY RULES

- Pre-oxygenate with 100% O₂.
- Insert catheter without suction; apply suction only during 360° rotation and withdrawal.
- Max time: **10 to 15 seconds!**



Tracheostomy Speed Check

What is the maximum safe pressure range for an indwelling tracheostomy cuff?

20 to 25 mmHg (or 25 to 30 cmH2O to prevent tracheal ischemia)

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DEAFNESS, HEARING AIDS & AURAL REHABILITATION

⚡ **HEARING IMPAIRMENT:** Conductive hearing loss (outer/middle ear block; corrected by amplification) vs. Sensorineural hearing loss (cochlear/8th nerve damage; permanent). **Hearing aid care requires cleaning with soft dry cloth—never immerse in water!**

HEARING LOSS TYPE	PATHOPHYSIOLOGICAL MECHANISM & AUDIOMETRIC FINDINGS	COMMON ETIOLOGY & MANAGEMENT
CONDUCTIVE HEARING LOSS	Impaired sound wave transmission through external or middle ear to cochlea. Bone conduction is better than air conduction (Negative Rinne).	Impacted cerumen, otitis media, otosclerosis, tympanic membrane perforation. Treated medically or surgically (stapedectomy).
SENSORINEURAL LOSS	Damage to cochlea (hair cells) or Vestibulocochlear Nerve (CN VIII). Air and bone conduction both impaired equally (Positive Rinne).	Presbycusis (age-related), noise-induced trauma, Meniere's disease, ototoxic drugs. Permanent; aided by cochlear implants.

HEARING AID CARE & COMMUNICATION STRATEGIES

HEARING AID MAINTENANCE

- Turn off aid and remove battery when not in use.
- Clean ear mold with mild soap and water; **never immerse electronic amplifier in water!**
- Clear cerumen from cannula using special pick.

COMMUNICATING WITH DEAF CLIENTS

- Face the client directly in good lighting; speak clearly in a **low-pitched, normal tone** (shouting high-pitched tones worsens sensorineural perception!).
- Eliminate background noise (turn off TV). Use written notes or gestures if needed.



Deafness Speed Check

How should the nurse modulate voice tone when speaking to a hearing-impaired client? **Speak in a lower-pitched, normal conversational tone directly facing the client**

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NUR-09 Batch 1: Units I, II, III & IV (Sensory & Renal)

 **EYE ASSESSMENT:** Evaluation of visual acuity and intraocular pressure. **Snellen chart tests distance vision at 20 feet**; Tonometry measures intraocular pressure (Normal IOP: 10–21 mmHg). Fundoscopy visualizes retinal vessels and optic disc.

OPHTHALMIC TEST	CLINICAL PROTOCOL & TESTING METHODOLOGY	NORMAL BASELINE & CLINICAL SIGNIFICANCE
SNELLEN VISUAL ACUITY	Client stands 20 feet (6 meters) from chart. Tests each eye individually with and without corrective lenses.	20/20 Vision: Client reads at 20 ft what normal eye reads at 20 ft. (20/200 = legally blind).
TONOMETRY (IOP Measurement)	Measures intraocular pressure using applanation tonometry or non-contact air-puff tonometer.	Normal IOP: 10 to 21 mmHg. Elevation (> 21 mmHg) indicates Glaucoma.
OPHTHALMOSCOPY	Direct visualization of internal ocular structures (retina, optic disc, macula, retinal vessels) in dimmed room.	Optic disc should be creamy pink/yellow with sharp margins (papilledema shows blurred margins).

PUPILLARY LIGHT REFLEX & EXTRAOCULAR MOVEMENTS

PUPILLARY REFLEX (PERRLA)

- Direct Light Reflex: Pupil constricts when light shines into it.
- Consensual Light Reflex: Opposite pupil constricts simultaneously.
- Accommodation: Pupils constrict when looking at near object.

EXTRAOCULAR MOVEMENTS (EOM)

- Tested via 6 Cardinal Fields of Gaze.
- Innervated by **Cranial Nerves III, IV, and VI** (SO4 LR6 All Else 3).

TONOMETRY CORNEAL ABRASION TRAP

When performing contact applanation tonometry, ensure topical anesthetic drops (Proparacaine/Tetracaine) are instilled first. Warn client not to rub their eyes post-test, as anesthetic numbing leaves the cornea vulnerable to **accidental scratching and corneal ulceration**.



Eye Assessment Speed Check

What is the normal physiological range for intraocular pressure (IOP)? **10 to 21 mmHg (Measured via tonometry)**

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REFRACTIVE ERRORS & EYELID/CONJUNCTIVAL DISORDERS

⚡ **REFRACTIVE & ADNEXAL DISORDERS:** **Myopia** (Nearsightedness; eyeball too long; corrected with Concave lenses); **Hyperopia** (Farsightedness; eyeball too short; Convex lenses). **Conjunctivitis** ("Pink Eye") is highly contagious!

REFRACTIVE ERROR	OPTICAL DEFECT & FOCAL POINT RELATIVE TO RETINA	CORRECTIVE LENS PRESCRIBED	CLINICAL PRESENTATION
MYOPIA (Nearsightedness)	Eyeball is too long OR cornea is too curved. Light rays focus IN FRONT OF RETINA .	Concave Lenses (diverging lenses).	Cannot see distant objects clearly; squints when looking far.
HYPEROPIA (Farsightedness)	Eyeball is too short OR cornea is too flat. Light rays focus BEHIND RETINA .	Convex Lenses (converging lenses).	Cannot see close objects clearly; eyestrain during reading.
ASTIGMATISM	Irregular curvature of cornea or lens causing light rays to focus on multiple focal points.	Cylindrical Lenses (corrects unequal corneal curvature).	Blurred or distorted vision at all distances.
PRESBYOPIA	Age-related loss of lens elasticity and ciliary muscle accommodation (starts around age 40–45).	Bifocals or Reading Glasses (Convex).	Needs longer arms to read books/menus.

EYELID INFLAMMATION & CONJUNCTIVITIS

HORDEOLUM VS CHALAZION

- • **Hordeolum (Stye):** Acute bacterial infection of hair follicle/sebaceous gland of eyelid margin. Painful red tender nodule. Warm compresses.
- • **Chalazion:** Chronic sterile granuloma of meibomian gland. painless hard nodule.

CONJUNCTIVITIS ("PINK EYE")

- • **Bacterial:** Purulent thick discharge, crusting in morning. Antibiotic drops.
- • **Viral:** Watery discharge, highly contagious. Strict hand hygiene.
- • **Allergic:** Itching, watery discharge, cobblestone papillae.



Refractive Errors Speed Check

What type of corrective lens is prescribed for Myopia (nearsightedness)?

Concave (diverging) lenses

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CORNEAL DISORDERS & OCULAR TRAUMA EMERGENCIES

 **CORNEAL ULCER & TRAUMA:** The cornea is avascular and highly sensitive. **Corneal ulcers are sight-threatening emergencies (risk of perforation)!** Chemical eye splashes require immediate continuous irrigation for at least 15–30 minutes.

OCULAR CONDITION	PATHOPHYSIOLOGICAL HALLMARK & CLINICAL SIGNS	EMERGENCY NURSING MANAGEMENT	
CORNEAL ULCER / KERATITIS	Infection of cornea (bacterial, viral HSV, fungal) destroying epithelial layer. Severe eye pain, photophobia, foreign body sensation, blurred vision, hypopyon (pus in anterior chamber).	• Topical broad-spectrum antibiotics or antivirals. • Cycloplegic eye drops (dilates pupil, paralyzes accommodation to reduce pain). • NEVER PATCH AN INFECTED CORNEA (dark warm environment accelerates bacterial/fungal proliferation!).	
CHEMICAL EYE SPLASH	Acid or alkali chemical contact (alkali burns are worse because they penetrate deeper and liquefy tissue!).	Intense burning pain, conjunctival injection, epithelial sloughing, corneal opacification.	• EMERGENCY IMMEDIATE IRRIGATION with normal saline or tap water for 15 to 30 minutes continuously before transport! Check pH.
PENETRATING OCULAR TRAUMA	Foreign body or sharp object piercing globe. Extrusion of intraocular contents.	Protruding tissue, irregular pupil, shallow anterior chamber, vision loss.	• DO NOT REMOVE EMBEDDED OBJECTS! • Apply rigid eye shield (never pressure patch!); keep client flat; urgent ophthalmology referral.

CHEMICAL BURN RULE

- Time is vision! Flush eye immediately with water or saline.
- Continue irrigation until conjunctival pH returns to normal neutral (pH 7.0–7.3).

EMBEDDED FOREIGN BODY TRAP

Pulling an embedded foreign body or sharp object out of an injured eye causes **intraocular contents (vitreous humor / iris) to prolapse and gush out**, causing permanent blindness. Stabilize with a rigid shield and leave removal to the surgeon!

 Corneal Disorders Speed Check

What is the immediate emergency nursing action for a chemical splash to the eye? **Immediate continuous copious irrigation with normal saline or water for 15–30 minutes**

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CATARACT PATHOPHYSIOLOGY & POST-OP IOL CARE

CATARACT PATHOPHYSIOLOGY: Opacification or clouding of the eye's crystalline lens. **Classic Manifestations:** Painless, progressive blurred vision, cloudy/milky pupil (leukocoria), halos around lights, impaired night driving. Treated via Phacoemulsification + IOL implant!

CATARACT ETIOLOGY	RISK FACTORS & PATHOLOGICAL MECHANISM	CLINICAL PRESENTATION
SENILE (AGE-RELATED)	Oxidative stress, protein denaturation, and clumping within lens over decades (> 60 yrs).	Gradual painless loss of visual acuity, faded color perception, frequent eyeglass prescription changes.
TRAUMATIC & METABOLIC	Blunt/penetrating eye trauma, chronic diabetes mellitus, long-term systemic corticosteroid therapy.	Rapidly developing opacification; rosette-shaped traumatic cataracts.
CONGENITAL	Maternal infections during first trimester (Rubella, Cytomegalovirus), genetic syndromes.	Absence of red reflex in neonate; cloudy lens (Leukocoria).

POST-OPERATIVE CATARACT SURGERY: INTRAOCULAR PRESSURE (IOP) PREVENTION

grid-2

ACTIVITIES STRICTLY PROHIBITED (RAISES IOP!)

- • **NO BENDING BEND WAIST** or lifting heavy objects (> 5 lbs).
- • NO straining during bowel movements (prescribe stool softeners!).
- • NO coughing, sneezing forcefully, or vomiting.
- • NO rubbing or pressing on the operated eye.

PROTECTIVE POSITIONING & SHIELD

- • Wear eye shield at night and protective sunglasses outdoors.
- • Sleep on the **unaffected side** or back (never sleep on operative side!).
- • Instill prescribed antibiotic and steroid eye drops aseptically.

POST-OP ACUTE PAIN / HEMORRHAGE TRAP

Mild scratchy sensation is normal post-cataract surgery, but **sudden sharp pain, severe eye ache, decreasing vision, or nausea/vomiting** indicates acute intraocular hemorrhage or acute angle-closure glaucoma. Notify surgeon stat!



Cataract Speed Check

What sleeping position is recommended following cataract surgery?

Sleep on the back or unaffected (non-operative) side to prevent pressure on operative eye

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GLAUCOMA MASTER: OPEN-ANGLE VS ANGLE-CLOSURE

GLAUCOMA MASTER: Optic neuropathy characterized by elevated Intraocular Pressure (> 21 mmHg) leading to optic disc cupping and peripheral vision loss. **Acute Angle-Closure Glaucoma is an ocular emergency requiring emergency miotics & osmotic diuretics!**

GLAUCOMA TYPE	PATHOPHYSIOLOGICAL MECHANISM	CLINICAL PRESENTATION	EMERGENCY TREATMENT
OPEN-ANGLE (POAG) (Most Common / Chronic)	Gradual outflow obstruction of aqueous humor through trabecular meshwork. Elevated IOP develops silently.	Slow, painless loss of peripheral vision ("Tunnel Vision") ; central vision remains intact until late stages. Often discovered during routine eye exam.	• Beta-blockers (Timolol drops). • Prostaglandin analogs (Latanoprost). • Carbonic anhydrase inhibitors.
ANGLE-CLOSURE (PACG) (Acute / Emergency)	In angle-closure glaucoma, the iris bulges forward and blocks the drainage angle, causing a rapid increase in IOP, often reaching > 30-40 mmHg.	Severe excruciating ocular pain (often described as halos around lights), redness, and decreased vision. Often associated with nausea and vomiting.	EMERGENCY IV MANNITOL (Osmotic diuretic) + Pilocarpine (Miotics)

MYDRIATIC (DILATING) DRUG CONTRAINDICATION IN GLAUCOMA

THE MYDRIATIC HAZARD

- Anticholinergic drugs (Atropine, Tropicamide, Scopolamine) cause pupil dilation (mydriasis).
- In angle-closure glaucoma, pupil dilation bunches the iris against the drainage angle, **completely sealing aqueous outflow and triggering an acute blindness crisis!**

EMERGENCY OSMOTIC THERAPY

- IV Mannitol (osmotic diuretic) rapidly draws fluid out of the vitreous humor and aqueous chamber into blood vessels, rapidly lowering IOP prior to surgery.



Glaucoma Speed Check

What medication is strictly contraindicated in patients with Glaucoma due to pupillary dilation?

Anticholinergic drugs (Atropine / Mydriatics)

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NUR-09 Batch 1: Units I, II, III & IV (Sensory & Renal)

RETINAL DETACHMENT & KERATOPLASTY GUIDELINES

 **RETINAL DETACHMENT:** Separation of sensory retina from underlying retinal pigment epithelium (RPE). **Classic Triad:** Sudden flashes of light (photopsia), shower of dark floaters, and "curtain falling" across visual field! Surgical emergency.

CONDITION / PROCEDURE	PATHOPHYSIOLOGICAL HALLMARK & CLINICAL SIGNS	CRITICAL NURSING & SURGICAL PROTOCOL
RETINAL DETACHMENT	Traction or fluid accumulation separates neural retina from blood supply. Painless vision loss starting in peripheral fields and progressing centrally.	• EMERGENCY RESTRICTION: Bed rest, patch both eyes to minimize eye movement. • Position head per surgeon's order (keeps gas bubble against tear). • Scleral buckling / vitrectomy.
KERATOPLASTY (Corneal Transplant)	Surgical replacement of diseased or scarred host cornea with a clear donor corneal button (penetrating keratoplasty).	• Post-op care: Wear eye shield at night, avoid bending/straining. • Monitor for graft rejection (RSVP: Redness, Sensitivity to light, Vision decrease, Pain).

EYE DONOR RETRIEVAL & CORPSE CARE PROTOCOL

EYE RETRIEVAL NURSING STEPS

- 1. Obtain legal family consent for eye donation promptly after death.
- 2. Elevate head of bed to **30–45 degrees** (prevents post-mortem corneal edema).
- 3. Apply antibiotic eye drops and gently close eyelids using paper tape.
- 4. Place small **ice packs over closed eyes** to preserve corneal tissue until harvest team arrives (within 4–6 hours).

CORNEAL GRAFT REJECTION SIGNS

- • Host immune system attacks donor corneal graft weeks to months post-op.
- • Watch for acute corneal clouding, ciliary flush, severe eye pain, and photophobia. Treated with topical steroid eye drops.



Retinal Detachment Speed Check

What visual symptom is classically described as a "curtain falling" over vision? **Retinal Detachment (Separation of sensory retina from pigment epithelium)**

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RENAL ASSESSMENT, URINALYSIS & DIAGNOSTICS

RENAL ASSESSMENT: Evaluation of urinary and kidney function. **Serum Creatinine (Normal: 0.6–1.2 mg/dL) is the gold standard kidney filtration marker;** BUN reflects hydration and protein catabolism; Intravenous Pyelogram (IVP) requires iodine allergy screening!

DIAGNOSTIC TEST	CLINICAL PURPOSE & PROCEDURAL PROTOCOL	CRITICAL NURSING PRECAUTION
SERUM CREATININE & BUN	Measures nitrogenous waste clearance. Creatinine is muscle waste (unaffected by diet); BUN reflects renal function + hydration status.	Elevated BUN with normal creatinine indicates prerenal dehydration rather than intrinsic renal failure!
INTRAVENOUS PYELOGRAM (IVP)	X-ray visualization of kidneys, ureters, and bladder following IV radiopaque contrast injection.	• Screen for iodine / shellfish allergy. • Check baseline serum creatinine (contrast is nephrotoxic!). • Enforce pre-op bowel prep and hydration.
CYSTOSCOPY	Direct endoscopic visualization of bladder lumen and urethral meatus via cystoscope.	• Post-op: Expect mild pink-tinged urine and dysuria for 24–48 hours. • Monitor for bright red bleeding or urinary retention.

PERCUTANEOUS RENAL BIOPSY NURSING CARE

PRE & POST-BIOPSY PROTOCOL

- Pre-op: Verify PT/INR, aPTT, and platelet counts (bleeding risk!). NPO 6–8 hours.
- **Post-op:** Place client in **PRONE position with towel roll under abdomen for 30–60 minutes** to provide natural pressure hemostasis!
- Maintain strict bed rest for 24 hours. Monitor vital signs q15m x 4.

POST-BIOPSY HEMORRHAGE SIGNS

- Kidneys are highly vascular; bleeding into renal capsule causes severe flank pain radiating to groin and gross hematuria.
- Notify surgeon stat; monitor for hypovolemic shock.



Renal Diagnostics Speed Check

What is the required client positioning immediately following a percutaneous renal biopsy?

Prone position with a towel roll under the abdomen for 30–60 minutes to provide pressure hemostasis

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LOWER & UPPER UTI & PYELONEPHRITIS

⚡ **URINARY TRACT INFECTIONS:** Lower UTI (Cystitis: Dysuria, frequency, urgency, suprapubic pain) vs upper UTI (Acute Pyelonephritis: **CVA tenderness, high fever, shaking chills, WBC casts in urine**). E. coli is #1 causative pathogen!

UTI CLASSIFICATION	ANATOMICAL SITE & PATHOGEN	CLINICAL PRESENTATION	NURSING PRIORITY
CYSTITIS (Lower UTI)	Infection of urinary bladder mucosa. Pathogen: *E. coli* (80% of cases).	Dysuria, urinary frequency, urgency, suprapubic discomfort, cloudy foul-smelling urine, hematuria.	Encourage high fluid intake (2–3 L/day), prescribe antibiotics (Nitrofurantoin/ Trimethoprim).
ACUTE PYELONEPHRITIS (Upper UTI)	Bacterial infection ascending from bladder into renal pelvis and renal parenchyma.	• Costovertebral Angle (CVA) tenderness. • Abrupt high fever, shaking chills, nausea, vomiting, flank pain. • WBC Casts in urinalysis.	Blood & urine cultures, IV broad-spectrum antibiotics (Ceftriaxone/ Fluoroquinolones), monitor renal function.

CATHETER-ASSOCIATED UTI (CAUTI) PREVENTION BUNDLE

CAUTI PREVENTION STANDARDS

- Insert indwelling catheters only when strictly indicated; remove as soon as clinically possible.
- Maintain closed sterile drainage system at all times.
- Keep urine collection bag **below bladder level**; never let bag touch floor.
- Perform routine perineal hygiene with soap and water daily.

CATHETER IRRIGATION TRAP

Routine prophylactic irrigation of indwelling urinary catheters with sterile water or antiseptics is **strictly prohibited**! Irrigation introduces bacteria and significantly increases the incidence of CAUTI.



UTI Speed Check

What urinary microscopic finding distinguishes upper pyelonephritis from lower cystitis?

White Blood Cell (WBC) Casts in urinalysis (Indicates renal parenchymal infection)

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GLOMERULONEPHRITIS VS NEPHROTIC SYNDROME

NEPHRITIC VS NEPHROTIC SYNDROME: Acute Glomerulonephritis (AGN: Post-streptococcal, hematuria / cola urine, hypertension, mild edema) vs Nephrotic Syndrome (Massive proteinuria > 3.5 g/day, severe hypoalbuminemia, anasarca, hyperlipidemia)!

RENAL SYNDROME	PATHOPHYSIOLOGICAL HALLMARK & TRIGGER	CLINICAL PRESENTATION	CORE NURSING CARE
ACUTE GLOMERULONEPHRITIS (AGN)	Immune complex deposition in glomerular basement membrane following Group A Beta-Hemolytic Strep infection (1–3 weeks post-pharyngitis/impetigo).	• Hematuria (Cola-colored / smoky urine). • Periorbital facial edema (worse in morning). • Hypertension (fluid overload). • Mild-to-moderate proteinuria (< 2 g/day).	• Throat cultures for strep. • Antibiotics (Penicillin). • Sodium and fluid restriction. • Antihypertensives & diuretics.
NEPHROTIC SYNDROME	Increased glomerular permeability to plasma proteins due to immune injury or podocyte damage. Massive loss of serum albumin in urine.	• Massive Proteinuria (> 3.5 g/24 hrs). • Severe Hypoalbuminemia (< 2.5 g/dL). • Generalized Anasarca (pitting edema, ascites, pleural effusion). • Hyperlipidemia & lipiduria (fatty casts).	• Low-sodium, moderate-protein diet. • Corticosteroids (Prednisone) to suppress immune response. • Diuretics & daily weights. • Monitor for thromboembolism (loss of antithrombin III).

EDEMA DISTRIBUTION & DIETARY MANAGEMENT

EDEMA PATTERNS

- **AGN:** Periorbital facial edema (especially noticeable upon waking in the morning) and mild dependent edema.
- **Nephrotic Syndrome:** Generalized severe "Anasarca" (swelling of face, abdomen/ascites, scrotal/labial edema, and lower limbs).

NUTRITIONAL THERAPY

- Nephrotic syndrome requires **moderate protein intake** (replacing urinary protein losses without accelerating renal burnout) combined with strict sodium restriction to control edema.



Glomerular Speed Check

What urinary finding is classically associated with Acute Post-Streptococcal Glomerulonephritis?

Hematuria resulting in smoky or cola-colored urine

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RENAL CALCULI, ESWL LITHOTRIPSY & URINE STRAINING

 **RENAL CALCULI (KIDNEY STONES):** Mineral deposits in renal pelvis/ureter. **Types:** Calcium Oxalate (most common), Uric Acid, Struvite (infection), Cystine. Classic symptom: Excruciating colicky flank pain radiating to groin (renal colic)!

STONE TYPE	ETIOLOGY & CONTRIBUTING FACTORS	RADIOLOGICAL VISIBILITY	PREVENTIVE DIETARY THERAPY
CALCIUM OXALATE (70%–80%)	Hypercalcemia, hyperoxaluria, dehydration, hyperparathyroidism.	Radiopaque (clearly visible on plain KUB X-ray).	Reduce dietary oxalate (spinach, nuts, tea, chocolate); adequate hydration; thiazide diuretics.
URIC ACID (5%–10%)	Gout, high purine diet (red meat, shellfish), acidic urine (pH < 5.5).	Radiolucent (invisible on plain X-ray; requires CT or ultrasound).	Low-purine diet; Alkalinize urine with Potassium Citrate; Allopurinol.
STRUVITE (Magnesium Ammonium Phosphate)	Chronic alkaline urinary tract infections caused by urea-splitting bacteria ("Proteus, Pseudomonas"). Forms "Staghorn calculi".	Radiopaque (large branching staghorn stones).	Treat underlying UTI; acidify urine; surgical removal.

EXTRACORPOREAL SHOCK WAVE LITHOTRIPSY (ESWL) & URINE STRAINING

ESWL POST-PROCEDURAL CARE

- Uses high-energy shock waves to shatter renal stones into tiny sand-like fragments.
- **Expected finding:** Mild bruising over flank and **gross hematuria** for 24–48 hours post-procedure.
- Encourage high fluid intake (3–4 L/day) to flush out stone debris.

URINE STRAINING PROTOCOL

- **MANDATORY NURSING ACTION:** Strain all client urine through a gauze/urine strainer until stone fragments are retrieved!
- Send retrieved stone fragments to laboratory for mineral composition analysis.



Renal Calculi Speed Check

What is the mandatory post-procedural nursing action following ESWL for kidney stones?

Strain all urine through gauze to retrieve stone fragments for laboratory analysis

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ACUTE KIDNEY INJURY (AKI), CKD & DIALYSIS INDICATIONS

🚨 **AKIT VS CKD:** Acute Kidney Injury is abrupt, potentially reversible renal failure (Prerenal, Intrarenal, Postrenal); Chronic Kidney Disease is progressive, irreversible nephron destruction. **Hyperkalemia is the most lethal acute complication of renal failure!**

AKI CATEGORY	PATHOPHYSIOLOGICAL MECHANISM & CAUSES	CLINICAL LABORATORY CLUES	PRIMARY MANAGEMENT
PRENAL AKI	Reduced renal blood flow / hypoperfusion (shock, hemorrhage, dehydration, heart failure, sepsis).	BUN:Cr ratio > 20:1; Urine sodium < 20 mEq/L; FENa < 1%. (Kidneys retain sodium to save water).	Rapid volume expansion (IV crystalloids/blood); restore cardiac output.
INTRARENAL AKI	Direct damage to renal parenchyma, glomeruli, or tubules (Acute Tubular Necrosis / ATN from nephrotoxic drugs/contrast, rhabdomyolysis).	BUN:Cr ratio ~ 10:1; Urine sodium > 40 mEq/L; Muddy brown casts in urine sediment.	Avoid nephrotoxic drugs (NSAIDs, aminoglycosides, contrast); adjust drug dosages to GFR.
POSTRENAL AKI	Mechanical urinary tract obstruction outflow (BPH, prostate cancer, bilateral ureteral calculi, strictures).	Anuria or alternating oliguria and polyuria; bilateral hydronephrosis on ultrasound.	Urological catheterization, ureteral stent placement, nephrostomy tube decompression.

DIALYSIS INDICATIONS (AEIOU MNEMONIC)

AEIOU EMERGENCY INDICATIONS

- **A = Acidosis:** Severe metabolic acidosis ($\text{pH} < 7.1$).
- **E = Electrolytes:** Severe Refractory Hyperkalemia ($\text{K}^+ > 6.5$ with ECG changes).
- **I = Ingestions:** Overdose of dialyzable toxins (Lithium, Aspirin, Alcohols).
- **O = Overload:** Fluid overload causing refractory pulmonary edema.
- **U = Uremia:** Encephalopathy, pericarditis, or bleeding diathesis.

OLIGURIC PHASE FLUID OVERLOAD TRAP

During the oliguric phase of AKI, urine output drops below 400 mL/day. Unmonitored IV fluid administration leads directly to **hypertensive crisis, acute heart failure, and fatal pulmonary edema!** Fluid intake must equal urine output plus 500 mL insensible loss.

🎯 AKI & CKD Speed Check

What mnemonic summarizes emergency clinical indications for immediate dialysis?

AEIOU (Acidosis, Electrolytes/Hyperkalemia, Ingestions, Overload, Uremia)

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■ **BURNS: Airway FIRST** — inhalation injury kills before fluids! Depth (1st/2nd/3rd/4th) + TBSA (Rule of Nines / Lund-Browder) decides Parkland + referral.

BURN DEPTH — BLANCH TEST + SENSATION

Depth	Layer	Look + Feel	Healing
1st (epidermal)	Epidermis only	Red, painful, blanches, NO blisters (sunburn)	7 days, no scar
2nd superficial-partial	Epidermis + upper dermis	Pink, moist BLISTERS, very painful, blanches	14–21 days, minimal scar
2nd deep-partial	Deep dermis	Mottled white-red, sluggish blanch, reduced pain	3–8 wks, graft often needed
3rd (full-thickness)	All dermis + appendages	White/charred/leathery, DRY, painless (nerves dead), NO blanch	Graft mandatory
4th degree	Beyond skin — fascia/muscle/bone	Black eschar, compartment risk	Amputation/reconstruction may be needed

RULE OF NINES (ADULT) + PALM + LUND-BROWDER

- Head/neck 9% | Each arm 9% | Anterior trunk 18% | Posterior trunk 18% | Each leg 18% | Perineum 1%.
- Child: head LARGER (18%), legs smaller — use Lund-Browder chart (most accurate, age-adjusted).
- Palm of patient hand (incl fingers) ≈ 1% TBSA — for scattered/small burns.
- Circumferential full-thickness limb/chest burn → escharotomy (limb perfusion / chest excursion). Inhalation signs: singed nasal hair, sooty sputum, hoarseness, stridor.

■ **FATAL TRAP**

Cooling a major burn with ice-cold water or prolonged immersion causes hypothermia + vasoconstriction and deepens tissue injury! Use room-temp sterile saline, cover with clean dry sheet, keep NPO/warm.

■ **NORCET HIT**

NORCET HIT: Rule of 9s — Adult head 9, each arm 9, trunk 36 (18+18), each leg 18, perineum 1. Blanch = partial thickness; NO blanch + painless = full thickness. Mnemonic: 1-PAIN red, 2-BLISTERS, 3-PAINLESS white.

■ **An adult has circumferential full-thickness burns of entire right arm + anterior trunk. TBSA?**

Right arm 9% + anterior trunk 18% = 27% TBSA. Circumferential → monitor perfusion, prepare escharotomy.

■ **PARKLAND: 4 mL × kg × %TBSA (2nd + 3rd degree only) — half in FIRST 8 hrs from BURN time, not admission! Urine output is the endpoint.**

PARKLAND FORMULA + ENDPOINTS

- Volume (24h) = 4 mL × weight (kg) × %TBSA (partial + full thickness; exclude 1st degree). Ringer Lactate preferred.
- HALF in first 8 hrs from time of burn, remaining half over next 16 hrs. Example: 70 kg × 40% = 11,200 mL/24h → 5,600 mL in first 8h (=700 mL/hr).
- Titrate to urine output: adult ≥0.5 mL/kg/hr (≥30 mL/hr); child ≥1 mL/kg/hr; + MAP ≥65, HR, mentation, lactate.
- Burn shock = hypovolemic + capillary leak (first 24–48h) → then hypermetabolic phase: high protein/calorie need, ileus, Curling ulcer risk.

Phase	Timing	Priority
Emergent	0–48h	ABCDE, airway, Parkland, Foley, NG, tetanus, escharotomy watch
Acute	48h–wound closure	Infection control, wound care, nutrition (high protein), pain, physio
Rehab	Months–years	Contracture prevention, pressure garments, psychosocial, graft care

■ FATAL TRAP

Calculating Parkland from admission time instead of burn time, or including 1st-degree area, over-resuscitates ('fluid creep' → compartment syndrome, ARDS)! Always ask burn time, count only 2nd/3rd degree.

■ NORCET HIT

NORCET HIT: 4 × kg × %TBSA — RL, 1/2 in 8h from BURN. Best resuscitation guide = URINE OUTPUT. Curling ulcer (duodenal stress ulcer post-burn) → PPI prophylaxis. Mnemonic: PARK — Percent, Adult-kilos, Ringer, Kidney-output.

■ 60-kg patient, 30% second-degree burns sustained 2 hrs ago. First-8-hour rate?

Total = 4×60×30 = 7,200 mL. Half (3,600 mL) over first 8h from burn → 450 mL/hr; subtract fluids already given.

■ **BURN WOUND: Clean → debride → topical antimicrobial → sterile dressing! Infection (Pseudomonas) + ileus + Curling ulcer are the silent killers.**

WOUND CARE LADDER + TOPICALS

Agent	Best for	Trap
Silver sulfadiazine 1%	Gram-neg incl Pseudomonas; painless	Transient leukopenia; avoid face/eyes; sulfa allergy
Mafenide acetate	Deep/full-thickness, eschar penetration	Painful; metabolic acidosis (carbonic anhydrase inhibitor)
Silver nitrate 0.5%	Large wounds	Stains black; leaches Na/K/Ca → monitor electrolytes
Honey / moist dressings	Superficial-partial	Do NOT rupture intact blisters (except large/over joint)

- Handwash + sterile technique; daily hydrotherapy only if not hypothermic; tetanus prophylaxis mandatory.
- Nutrition: HIGH protein (1.5–2 g/kg) + HIGH calories + Vit A/C/zinc; early enteral feeding prevents ileus/translocation.
- Sepsis flags: fever spike, tachycardia, ileus, cloudy/thick odour wound, falling platelets → culture + escalate.

■ **FATAL TRAP**

Applying greasy ointments, butter, ice or toothpaste on fresh burns traps heat and seeds infection! Cool with room-temp saline, remove jewellery/constrictors, cover with sterile non-adherent dressing.

■ **NORCET HIT**

NORCET HIT: Pseudomonas = green-blue pus + fruity/grape odour. Curling ulcer → antacid/PPI. Position: elevate burned limbs; neck burns — NO pillow (prevent flexion contracture). Splints keep joints in FUNCTION.

■ **Most feared early GI complication after major burns and its prophylaxis?**

Curling ulcer (acute duodenal stress ulcer) → PPI/H2-blocker + early enteral nutrition.

GRAFT: Autograft (self) is gold standard — immobilize + NO shear! Graft failure = pale/cool + no capillary refill; donor site = moist + painful.

GRAFT TYPES + VIABILITY

Type	Source	Key point
Split-thickness (STSG)	Epidermis + part dermis; donor regrows	Large cover; meshed; donor painful pink
Full-thickness (FTSG)	All dermis; face/hands	Better cosmetic; donor needs closure
Autograft	Self	No rejection; best take
Allograft / Xenograft	Donor / animal (pig)	Temporary biologic cover only
Flap	Skin + vessels/muscle	Needs patent pedicle; check Doppler

- Protect graft 3–5 days: immobilize, elevate above heart, NO pressure/shear; report hematoma/seroma under graft (lifts graft → necrosis).
- Donor site: moist, very painful — fine-mesh gauze, keep dry, heals 10–14 days; infection = conversion to deeper wound.
- Rehab: pressure garments 23h/day after healing, silicone sheets, active ROM, anti-contracture positioning (axilla abducted, mouth splint for facial).

■ **FATAL TRAP**

Early ambulation with dangling or shearing a fresh graft dislodges it and causes hematoma → graft death! Immobilize per protocol, keep joints extended/splinted, dangle only after surgeon clears.

■ **NORCET HIT**

NORCET HIT: Graft take signs — pink + warm + brisk refill Day 3–5. Failure — white/blue + cool + no refill + foul odour. Donor heals by re-epithelialization (keep CLEAN-DRY).

■ **Day-3 graft is pale, cool, with dark fluid collection beneath. Action?**

Graft failure/hematoma — notify surgeon immediately; do NOT aspirate/compress; keep immobilized and sterile.

NPWT (VAC): -75 to -125 mmHg sealed suction — boosts granulation, removes exudate. NEVER place foam on exposed vessels/nerves/malignancy.

NPWT PROTOCOL + DRESSING MAP

Dressing	Use	Change
Transparent film	Superficial, low exudate	3–7 days
Hydrocolloid	Light-moderate exudate, autolytic debridement	3–5 days; forms gel — normal
Foam + NPWT	Deep, heavy exudate, graft beds	NPWT 48–72h; foam sterile
Alginate	Bleeding/heavy exudate	Daily; needs secondary cover

- NPWT checklist: airtight seal, canister < full, tubing unkinked, pressure ordered, pain pre-medicated; alarm = leak/blockage.
- Stop NPWT + call surgeon for: bright-red bleed >100 mL, exposed artery, necrotic black expansion, severe pain.
- Pressure-injury link: Braden scale; 2-hourly turns, heel offloading, head-end ≤30° unless contraindicated.

■ **FATAL TRAP**

Continuing NPWT over untreated osteomyelitis, malignancy or exposed major vessels erodes the vessel → catastrophic bleed! Inspect wound bed each change; protect vessels with non-adherent layer.

■ **NORCET HIT**

NORCET HIT: NPWT pressure -75 to -125 mmHg. Foam change 2–3 days. Rising pain + fever + periwound erythema = infection. Braden ≤18 = at risk.

■ **NPWT alarm + loss of suction hissing at drape edge. First action?**

Check seal/leak — reinforce drape, straighten tubing, empty canister; if bleed or severe pain, STOP and escalate.

COSMETIC: Sterile + consent + photos + realistic expectations! Liposuction danger = fat embolism + fluid shift; laser danger = eye + burn.

PROCEDURE — RISK — NURSING

Procedure	Top risk	Nursing must
Laser resurfacing	Burns, pigment change, eye injury	Eye shields, smoke evacuator, sunscreen, no picking
Liposuction	Fat embolism, bleeding, fluid shift	Compression garment, I/O, watch dyspnea/chest pain
Abdominoplasty	Hematoma, DVT	Jackson-Pratt care, binder, early leg mobilization
Chemical peel	Scarring, infection	Neutralize per protocol, moisturize, sun block

- Pre: informed consent, allergy/drug history (anticoagulants, isotretinoin), smoking cessation, baseline photos.
- Post: compression as ordered, drain care (empty + measure), DVT prophylaxis, sun protection 6–12 months.
- Red flags: expanding hematoma, dusky flap, fever/pus, calf pain, chest pain/dyspnea → emergency.

■ **FATAL TRAP**

Massaging a fresh hematoma or removing compression early after lipo/abdominoplasty worsens bleeding and contour deformity! Keep garment on, measure drains, report rapid swelling/pallor.

■ **NORCET HIT**

NORCET HIT: Fat embolism triad — dyspnea + confusion + petechiae (24–72h post lipo/long-bone). Laser needs wavelength-matched eye protection for patient + staff.

■ **24h post-liposuction: sudden dyspnea, confusion, axillary petechiae. Suspect?**

Fat embolism syndrome — high-flow O₂, notify surgeon, prepare ICU transfer.

AFFIRMING CARE: Chosen name + pronouns + privacy + dignity! Watch neovagina dilation adherence, urethral/flap perfusion, VTE and mental-health support.

PERIOP NURSING PRIORITIES

- Respect: ask + document chosen name/pronouns; private room/exam; drape to expose only operative field; zero misgendering.
- Vaginoplasty: dilation schedule is graft survival — teach aseptic dilation, depth/width log, water-based lubricant, douching per protocol.
- Phalloplasty/metoidioplasty + chest surgery: flap checks (colour/warmth/refill/Doppler), binder/drain care, urinary catheter patency.
- Hormones: estrogen → VTE risk (mobilize, stockings); testosterone → polycythemia/acne/BP → labs/BP.

Red flag	Action
Foul discharge + fever post-vaginoplasty	Infection/necrosis — culture + surgeon call
Flap pale/blue, no Doppler	Vascular compromise — emergency
Urinary retention / no output	Catheter kink vs stricture — assess + escalate
Severe dysphoria / self-harm ideation	Stay, safety plan, psych liaison + family support

■ FATAL TRAP

Skipping dilation or using non-sterile technique after vaginoplasty causes stenosis + infection and loss of depth! Reinforce schedule, handwash, correct dilator size progression, follow-up.

■ NORCET HIT

NORCET HIT: Dilation keeps neovaginal depth (lifelong initially). VTE prophylaxis after any major affirming surgery. Consent must be informed + voluntary + mental-health assessed.

■ Post-vaginoplasty Day-5: missed dilations + foul odour + fever. Priority?

Infection/stenosis risk — vital signs, wound exam, notify surgeon, reinforce aseptic dilation adherence.

■ **GCS: E4 V5 M6 = 15 normal; ≤8 = coma — intubate! Unequal/dilated pupil + Cushing triad = herniation — escalate NOW.**

GLASGOW COMA SCALE + PUPILS

Component	Score	Test
Eye (E)	4 spont / 3 voice / 2 pain / 1 none	Open + track
Verbal (V)	5 oriented / 4 confused / 3 words / 2 sounds / 1 none	Person-place-time
Motor (M)	6 obeys / 5 localizes / 4 withdraws / 3 flexion / 2 extension / 1 none	Trapezius pressure; best limb
Pupils	Size + equal + reactive	Penlight; 2–3 mm, brisk

- Worst motor: decorticate flexion (cortical, arms flexed) vs decerebrate extension (brainstem, arms extended) — extension = worse.
- LP prep: fetal lateral or sitting bent; empty bladder; consent + platelets/ICP check; post — flat 4–6h, headache/hydration watch.
- Never LP with papilledema/focal deficit/GCS drop until raised ICP excluded → herniation risk. Kernig/Brudzinski + nuchal rigidity = meningeal irritation.

■ FATAL TRAP

Doing deep suctioning, LP or supine-flat positioning in raised ICP spikes pressure and precipitates herniation! Head-end 30°, neck neutral, avoid coughing/straining/suction >10 sec, stool softeners.

■ NORCET HIT

NORCET HIT: GCS ≤8 intubate. Cushing = rising BP (wide pulse) + bradycardia + irregular breathing. Dilated FIXED pupil = uncal herniation. CSF normal: clear, 70–180 mmH₂O, protein 15–45.

■ E2 V3 M4 + left pupil 5 mm non-reactive after head injury. GCS and action?

GCS 9 (E2+V3+M4) + blown pupil = herniation — head-end 30°, O₂, neurosurgery + CT STAT.

HEAD INJURY: Assume C-spine injury! Battle sign + raccoon eyes + CSF rhinorrhea = BASE skull fracture — NO nose-blowing, NO NG tube via nose!

INJURY MAP + HALO TEST

Type	Clue	Care
Concussion	Brief LOC, amnesia, headache	Observe 24h, neuro checks, no driving/alcohol
Contusion	Prolonged deficit, CT bleed	Neuro ICU, ICP watch
Coup-contrecoup	Impact + opposite-side injury	High ICP suspicion
Base skull fracture	Battle/raccoon, CSF leak, hemotympanum	Head-end 30°, sterile pad, no nose-blow/NG-nose
Depressed fracture	Palpable step, dura tear risk	Surgery + seizure prophylaxis

- CSF leak test: halo/ring sign on filter paper (blood centre + clear ring); glucose-positive; salty taste rhinorrhea.
- Monitoring: q15min neuro + GCS + pupils + vitals initially; worsening headache/vomiting/anisocoria = bleed expansion.
- Seizure, vomiting, unequal pupils, Cushing triad → CT + neurosurgery; keep NPO if GCS low.

■ FATAL TRAP

Inserting NG tube via nose or asking patient to blow nose with CSF rhinorrhea drives bacteria into meninges → meningitis! Test leak, loose sterile dressing, prophylactic antibiotics per order, HOB 30°.

■ NORCET HIT

NORCET HIT: Battle (mastoid bruise) + Raccoon (periorbital) = base fracture. Halo = CSF. Lucid interval (talk then deteriorate) = extradural hematoma — burr-hole emergency.

■ Fall + clear nasal drip forming halo + mastoid bruising. Diagnosis + 2 prohibitions?

Base skull fracture with CSF leak — do NOT blow nose, do NOT pass NG via nose.

■ ICP normal 5–15 mmHg; CPP = MAP – ICP (keep 60–70). Cushing triad + blown pupil = herniation — mannitol + hyperventilate + neurosurgery!

MONRO-KELLIE + CPP MATH

- Skull is rigid: blood + brain + CSF — rise in one raises ICP. Early: headache, vomiting (projectile), papilledema, sunset eyes.
- Late (Cushing): systolic ↑ + widening pulse pressure, bradycardia, irregular (Cheyne-Stokes) breathing.
- CPP = MAP – ICP. Example MAP 90 – ICP 20 = CPP 70 (good). CPP <60 = brain ischemia!
- Care bundle: HOB 30°, neck midline (remove tight collar/ties), quiet/dark, temp + glucose normal, stool soft, mannitol/3% saline + furosemide per order.

Do	Don't
HOB 30°, O2, normocapnia	Flat supine, neck flexion
Sedation + pain control	Fever, hypercapnia, hypoxia
Osmotherapy + drain per order	Prolonged suctioning/Valsalva
Hourly GCS + pupils + MAP	Fluid overload / hypotension

■ FATAL TRAP

Hyperventilating routinely or giving hypotonic fluids (D5W) in raised ICP worsens cerebral edema and ischemia! Maintain normocapnia (PaCO₂ 35–40) unless herniation — then brief hyperventilation as bridge.

■ NORCET HIT

NORCET HIT: Mannitol needs Foley (osmotic diuresis) + electrolytes. Fixed dilated = uncal herniation. Target: ICP <20, CPP 60–70, MAP ≥80, PaO₂ >80.

■ MAP 85, ICP 30. CPP and interpretation?

CPP = 85–30 = 55 mmHg — LOW (ischemia) — lower ICP + raise MAP per protocol.

■ **SCI: Do NOT move/log-roll alone! T6+ lesion + pounding headache + sweating above + HTN = AUTONOMIC DYSREFLEXIA — sit up + find trigger!**

LEVEL READS FUNCTION + SHOCK TYPES

Level	Function left	Meaning
C1–C3	Diaphragm lost	Ventilator-dependent
C4–C5	Shrug/biceps	Wheelchair, needs aids
C6–T1	Wrist/hand	Feeding independence varies
T6+	Autonomic dysreflexia risk	Bladder/bowel trigger watch
Below L1	Cauda equina, flaccid	Bowel/bladder areflexic

- Spinal shock: flaccid + absent reflexes + ileus + hypotension + bradycardia (hours–weeks) — then spasticity returns.
- Neurogenic shock: warm-dry-hypotensive-bradycardic (loss of sympathetic tone) → fluids + vasopressors, atropine ready.
- Dysreflexia triggers: full bladder (kinked Foley #1), constipation/impaction, tight clothes, pressure injury — pounding headache + flushing/sweat ABOVE + pale/cool BELOW + HTN + bradycardia.

■ **FATAL TRAP**

Laying flat a dysreflexic patient or ignoring a blocked Foley lets BP surge to stroke! SIT UP 90°, loosen constriction, check catheter (unkink/irrigate/replace), bowel, skin — antihypertensive per order, call rapid response.

■ **NORCET HIT**

NORCET HIT: Steroids (methylprednisolone) only <8h per protocol now debated — know window. Halo traction pins — torque + pin-site care. Log-roll with 3–4 staff, cervical collar always.

■ **T4 paraplegic + blocked Foley + BP 210/120 + headache + sweating forehead. First 2 actions?**

Sit upright 90° + relieve bladder obstruction (unkink/flush/replace catheter) + monitor BP.

DISC: Sciatica + dermatome numbness + weak dorsiflexion! Log-roll + firm mattress + laminectomy precautions — NO bending/lifting/twisting.

DISC vs COMPRESSION

Feature	Herniated disc	Cord compression (tumor/trauma)
Pain	Radiating leg (sciatica), worse cough/sneeze	Back pain + progressive weakness
Neuro	Dermatomal numbness, foot drop	Saddle anesthesia, bladder/bowel change
Test	Straight-leg raise +, MRI	MRI emergency
Care	Flat/firm, NSAIDs, physio, traction	Steroids + decompress ASAP

- Conservative: flat bed + log-roll, knee flex pillow, heat/cold, NSAIDs + muscle relaxant, weight + posture teaching.
- Laminectomy/discectomy: pre — consent + neuro chart + bowel/bladder; post — log-roll, wound/drain check, urge to void watch (retention), DVT prophylaxis.
- BLT ban 6–12 wks: no Bending, Lifting >5 kg, Twisting; sit only on straight chair, use log-roll to get up.

■ FATAL TRAP

Sitting cross-legged or bending to tie shoes early after laminectomy herniates the repair! Teach log-roll, raised toilet seat, reacher, and report CSF leak (clear wound drainage) + saddle numbness + incontinence instantly.

■ NORCET HIT

NORCET HIT: L4–L5/S1 = common lumbar herniation — foot drop + absent ankle jerk. Cervical — hand clumsiness + gait change. Cauda equina = surgical emergency.

■ Day-1 post-laminectomy wants to bend to pick slippers. Correct teaching?

Do NOT bend — call for help, log-roll to side, use reacher; follow BLT restrictions.

■ **SAH: Thunderclap headache ('worst of life') + nuchal rigidity + photophobia! Rebleed + vasospasm (Day 4–14) kill — nimodipine + quiet-dark room.**

SAH TRIAD + GRADING

- Berry aneurysm (anterior circle of Willis) rupture → bloody CSF, xanthochromia; CT + angiography confirm.
- Hunt-Hess: 1 headache → 5 coma/posturing; higher = worse. Sentinel leak headache days before major bleed.
- Vasospasm peak Day 4–14: new deficit + low flow → nimodipine, triple-H now modified (euvolemia + induced HTN per order), no hypotension.
- Care: absolute bed rest, dark-quiet single room, HOB 30°, stool softener, no straining/coughing/NSAIDs that raise bleed, SBP per target (often <140–160 pre-secure).

Drug	Role	Watch
Nimodipine	Prevent vasospasm	Hypotension; give on time 21 days
Mannitol / 3% saline	ICP control	Na + output
Antiepileptic	Seizure prophylaxis	Sedation
Stool softener + antiemetic	No Valsalva	Falls + Na

■ FATAL TRAP

Giving anticoagulants or letting SAH patient strain on bedpan re-ruptures the aneurysm! Bed rest, laxatives, dim lights, visitors limited, pain + anxiety controlled, aneurysm clipped/coiled ASAP.

■ NORCET HIT

NORCET HIT: Thunderclap + stiff neck + Kernig = SAH until proven. Coil vs clip. Rebleed peaks first 24h; vasospasm Day 4–14. Avoid lumbar puncture if mass/ICP signs.

■ Day-6 post-SAH: new hemiparesis + drowsy. Likely cause + drug?

Cerebral vasospasm — ensure nimodipine on schedule, euvolemia, notify neurosurgery STAT.

MENINGITIS: Fever + headache + stiff neck + Kernig/Brudzinski! Bacterial = emergency ceftriaxone + isolation-droplet + LP (if safe) — do NOT delay antibiotics for CT.

BACTERIAL vs VIRAL vs NCC

Feature	Bacterial	Viral	Neurocysticercosis
Onset	Abrupt, toxic	Gradual, flu-like	Seizures, chronic
CSF	Cloudy, high protein, LOW glucose, neutrophils	Clear, normal glucose, lymphocytes	Eosinophils, cysts on CT/MRI
Drug	Ceftriaxone + vancomycin ASAP	Acyclovir if HSV	Albendazole + steroids + antiepileptic
Prevent	Vaccine + droplet precaution	Supportive	Pig pork hygiene, wash produce

- Triage: ABCs + blood cultures + antibiotics within 1h; droplet mask until 24h of antibiotics (meningococcal).
- Encephalitis: altered sensorium + seizures + focal deficit (temporal HSV) → acyclovir early, ICP watch.
- NCC: single calcified/pork tapeworm larval cyst is top seizure cause in India — pork + open defecation link; health education.

■ **FATAL TRAP**

Delaying antibiotics to wait for CT/LP in suspected bacterial meningitis kills! Draw cultures, start empiric ceftriaxone immediately, then image/LP when stable — plus droplet isolation and chemoprophylaxis for contacts.

■ **NORCET HIT**

NORCET HIT: Brudzinski (neck flex → hip/knee flex) + Kernig (knee extend pain). Waterhouse-Friderichsen (meningococcal + adrenal bleed + purpura) = emergency hydrocortisone.

■ **Fever + stiff neck + cloudy CSF with low glucose + neutrophils. Organism type + first drug?**

Bacterial meningitis — ceftriaxone (+ vancomycin) within 1 hour after cultures.

SEIZURE: Time it + side-lying + NOTHING in mouth! Status (>5 min / back-to-back) = lorazepam + airway — hypoxia + acidosis kill neurons.

TYPE — CLUE — DRUG

Type	Clue	First drug
Generalized tonic-clonic	Cry + stiff + jerk + incontinence + postictal sleep	Valproate / levetiracetam
Absence	Blank stare, school-age, hyperventilation triggers	Ethosuximide
Myoclonic	Morning jerks	Valproate
Focal	One-sided march, aura	Carbamazepine / phenytoin
Status epilepticus	≥5 min or no regain	IV lorazepam → fosphenytoin

- During: side-lying, pillow under head, loosen neck, suction ready, O2, time + video; after: vitals + glucose + injury check.
- Phenytoin: therapeutic 10–20 mcg/mL; gum hyperplasia + hirsutism + zero-order kinetics; fosphenytoin safer IV (less necrosis).
- Counsel: adherence + sleep + no alcohol/flashing lights; driving/swimming restrictions; pregnancy — folate + teratogenicity talk (valproate).

■ **FATAL TRAP**

Forcing spoon/fingers into mouth or restraining limbs during seizure breaks teeth/jaw and dislocates joints! Turn side-lying, protect head, loosen clothes, give O2 + benzodiazepine per order, never leave alone.

■ **NORCET HIT**

NORCET HIT: Status = ABCs + lorazepam + glucose + ABG. Postictal Todd paresis mimics stroke (resolves hours). Ethosuximide = absence; phenytoin gums grow.

■ **Tonic-clonic ongoing 6 min, SpO2 88%. First 3 actions?**

Call help + side-lying + O2/airway + IV lorazepam + prepare fosphenytoin infusion.

STROKE: BE-FAST — alteplase <4.5h (ischemic only)! BP permissive high pre-lysis; swallow screen before ANY oral — aspiration kills.

ISCHEMIC vs HEMORRHAGIC

Feature	Ischemic (85%)	Hemorrhagic
Cause	Thrombus/embolus, AF	HTN bleed, aneurysm, AVM
CT	Normal early, infarct later	Hyperdense bleed at once
BP	Permissive (<220/120 if no lysis)	Lower aggressively per target
Clot buster	Alteplase <4.5h if eligible	CONTRAINDICATED — surgery/coil

- BE-FAST: Balance, Eyes, Face droop, Arm drift, Speech, Time (= brain). Last-known-well time decides lysis/thrombectomy (<24h large vessel).
- Lysis checklist: glucose normal, no bleed/trauma/surgery 3 months, BP <185/110, NIHSS, consent; monitor for angioedema + bleed.
- Hemiplegia: affected side supported (pillow/sling, NO pulling arm), 2-hourly turns, swallow screen (GUSS), thickened fluids, Foleys avoided, DVT stockings, physio <24h.

■ FATAL TRAP

Giving aspirin/heparin before ruling out bleed, or feeding before swallow screen, causes fatal bleed/aspiration! CT first, NPO until screen pass, elevate HOB 30°, suction ready, speech therapy early.

■ NORCET HIT

NORCET HIT: Left MCA = aphasia + right weakness; Right = neglect + left weakness. Homonymous hemianopia — approach from intact side. Alteplase window 4.5h; thrombectomy up to 24h.

■ 2h sudden right weakness + aphasia, glucose 110, BP 170/95, CT no bleed. Eligible?

Likely ischemic left MCA — activate stroke code for alteplase/thrombectomy evaluation STAT.

BELL = LMN facial droop (CAN'T close eye + drool) — steroids + eye patch! TN = lightning face pain — carbamazepine + avoid trigger touch.

BELL (VII) vs TRIGEMINAL (V)

Feature	Bell palsy (VII)	Trigeminal neuralgia (V)
Lesion	LMN facial — whole half incl forehead	Sensory V2/V3 lightning pain
Clue	Can't close eye, drool, taste lost	Brush/teeth/shave triggers, no deficit
Drug	Prednisolone + acyclovir early	Carbamazepine first-line
Care	Eye patch + tears + face exercises	Soft food, oral hygiene, microvascular decompression option

- Bell eye: artificial tears day + ointment/patch night, sunglasses, facial sling/exercises, warn incomplete recovery + synkinesis.
- TN: small frequent lukewarm soft meals, chew on pain-free side, avoid cold wind/touch, monitor carbamazepine (drowsy + hyponatremia + marrow).
- Stroke vs Bell: stroke spares forehead (UMN — frontalis bilateral innervation); Bell involves forehead.

■ **FATAL TRAP**

Leaving Bell eye open without patch/tears ulcerates cornea → blindness! Tape/patch at night, drops by day, report eye pain/redness; steroids must start <72h for best recovery.

■ **NORCET HIT**

NORCET HIT: Bell = LMN (forehead involved); Stroke = UMN (forehead spared). TN trigger zones + carbamazepine + HLA-B*1502 Stevens-Johnson watch.

■ **Right face droop + cannot wrinkle forehead + cannot close eye. Bell or stroke?**

Bell palsy (LMN — forehead involved) — eye protection + steroids early.

NEUROPATHY: Glove-stocking numbness + foot drop + absent ankle jerk! Gabapentin + foot care + fall-proofing — monofilament yearly.

CAUSE — PATTERN — CARE

Cause	Pattern	Action
Diabetes	Stocking-glove, burning	HbA1c + gabapentin + foot exam
B12 deficiency	Dorsal column loss, ataxia	B12 IM + diet
Guillain-Barré (preview)	Ascending weakness	ICU + IVIG (next batch)
Drug (INH/vincristine)	Length-dependent	Pyridoxine + review drug

- Foot protocol: daily inspect (mirror), lukewarm water test by elbow, cotton socks, no barefoot, trim nails straight, callus debridement by clinic.
- Fall safety: night lights, walker, ankle-foot orthosis for foot drop, driving review.
- Pain ladder: duloxetine/gabapentin/pregabalin; avoid opioids first-line; glucose + alcohol control.

■ FATAL TRAP

Soaking feet in hot water or walking barefoot with insensate feet scalds/cuts unnoticed → ulcer + amputation! Test water with elbow, never barefoot, break-in shoes slowly, report blister/redness same day.

■ NORCET HIT

NORCET HIT: 10-g monofilament + 128-Hz tuning fork = screening. Absent ankle jerk + stocking pain = diabetic polyneuropathy. INH neuropathy → pyridoxine (B6).

■ Diabetic + burning feet + absent ankle jerks + small blister unnoticed. Priority teaching?

Daily foot inspection + no barefoot + elbow-test water + same-day report of any break.

■ **PARKINSON: TRAP tetrad** — Tremor (pill-rolling), Rigidity (cogwheel), Akinesia/bradykinesia, Postural instability! Levodopa-carbidopa is gold; protein blocks absorption.

TRAP + ON-OFF + DRUGS

Feature	Parkinson clue	Nursing
Tremor	Pill-rolling at rest, disappears with action	Weighted utensils, spill-proof
Rigidity	Cogwheel + lead-pipe	ROM, warm packs, fall watch
Akinesia	Shuffling, festination, freezing, mask face	Cue steps ('march'), wide turns
Posture	Stooped, retropulsion, falls	Walker, no rushing

- Levodopa-carbidopa: give 30 min before protein meals; ON (good move) vs OFF (freeze) vs dyskinesia (too much).
- Anticholinergics (trihexyphenidyl): dry mouth + retention + confusion in elderly — monitor. Selegiline + meperidine = fatal serotonin syndrome.
- Care: thickened fluids if dysphagia, upright after meals, constipation regimen, depression screen, DBS post-op neuro checks.

■ **FATAL TRAP**

Stopping levodopa abruptly or giving protein-rich meal with dose triggers neuroleptic-malignant-like rigidity + prolonged OFF freeze with aspiration risk! Taper only, time doses, small frequent low-protein-day meals.

■ **NORCET HIT**

NORCET HIT: Mask face + micrographia + shuffling + pill-rolling = Parkinson. Freezing → step over line/laser cue. Mnemonic TRAP. Falls = #1 killer.

■ **Parkinson on levodopa freezes 1h before next dose with festination. Phenomenon + fix?**

Wearing-OFF — notify for dose/frequency review; cueing + fall precautions meanwhile.

ALZHEIMER: 7As — Amnesia, Aphasia, Apraxia, Agnosia, Anomia, Abstract-loss, Attention-loss! Early short-term loss; late incontinence + wandering. Reorient, never argue.

STAGES + SUNDOWNING

Stage	Clue	Care
Mild	Short-term loss, repeats, misplaces	Lists, pill box, routines
Moderate	Sundowning, wandering, agnosia	Locked doors, ID band, night light
Severe	Incontinence, mutism, dysphagia	Puree, skin, aspiration watch

- Drugs: donepezil/rivastigmine (cholinesterase — bradycardia + GI upset); memantine moderate-severe; NO routine antipsychotics (stroke risk).
- Sundowning (evening agitation): daylight activity, low noise, toileting + pain check before sedating; validation, not confrontation.
- Safety: remove knobs/locks hazards, stove off, drive-stop talk, caregiver respite + support groups.

■ FATAL TRAP

Arguing, quizzing or restraining a sundowning elder escalates to violence and elopement! Redirect + calm room + offer snack/toilet/pain relief; restraints only last-resort with order + checks.

■ NORCET HIT

NORCET HIT: Reversible mimics — B12, hypothyroid, depression (pseudodementia), UTI, drugs. MMSE <24 impaired. First change = short-term memory.

■ Elder repeats questions, wanders at dusk, MMSE 19. Stage + first intervention?

Moderate Alzheimer — structured routine + wandering safety + caregiver plan.

■ **GBS: Post-viral ASCENDING weakness (feet → arms) + areflexia! Autonomic storm + respiratory failure kill — monitor VC/NIF + ICU + IVIG/plasmapheresis.**

MILLER-FISHER + MONITOR

Sign	GBS	Mimic (polio/transverse)
Pattern	Symmetric ascending	Asymmetric / level-based
Reflexes	Absent globally	Hyper below level (myelitis)
CSF	Albuminocytologic dissociation (high protein, normal cells)	Pleocytosis in infection
Trigger	Campylobacter/flu 1–3 wks prior	Vaccine/tick

- ICU: q1h vitals + continuous ECG/BP (autonomic lability), VC + NIF + gag/cough; intubate if VC <15 mL/kg or negative force weak.
- Treat: IVIG or plasmapheresis; NO steroids (ineffective); heparin + physio (DVT), eye + pressure care if ventilated.
- Recovery weeks–months, ascending reverses descending; physio + depression + pain (gabapentin) + vaccine counselling.

■ FATAL TRAP

Missing falling vital capacity while GBS patient 'talks fine' leads to sudden arrest! Track trends + bulbar signs (nasal voice, choking); keep tracheostomy tray + suction at bedside.

■ NORCET HIT

NORCET HIT: GBS = post-Campylobacter ascending areflexic weakness + high CSF protein/normal cells. Plasmapheresis/IVIG <4 wks. Worst days 2–4.

■ 2 wks post-diarrhea: leg weakness rising to hands + absent reflexes + VC falling. Action?

Suspected GBS — ICU, neuro exam + LP/MRI, IVIG workup, vent readiness.

MG: Anti-AChR antibodies — weakness WORSENS with activity, IMPROVES with rest! Ptosis + diplopia evenings; edrophonium improves. Cholinergic vs myasthenic crisis = opposite drugs!

CRISIS TABLE (EXAM FAVOURITE)

Feature	Myasthenic (too little drug)	Cholinergic (too much drug)
Cause	Missed dose / infection	Overdose pyridostigmine
Weakness	Severe, dysphagia, ptosis	Severe + fasciculations
Secretions	Normal/dry	SLUDGE: saliva, diarrhea, miosis
Edrophonium test	IMPROVES	WORSENS
Treat	More anticholinesterase + vent	STOP drug + atropine + vent

- Pyridostigmine: with milk/food, on-time (30 min before meals); report diarrhea/cramps/miosis = overdose.
- Thymectomy helps; avoid beta-blockers, aminoglycosides, quinolones (worsen weakness); plasmapheresis/IVIG for crisis.
- Tensilon test only with atropine + resuscitation ready (bradycardia risk); ice-pack test bedside for ptosis.

■ **FATAL TRAP**

Giving extra pyridostigmine to a cholinergic-crisis patient in respiratory failure deepens SLUDGE + arrest! Hold drug, suction, atropine + vent per order — edrophonium test differentiates first.

■ **NORCET HIT**

NORCET HIT: MG worse evening/activity, better rest/morning. Ptosis + dysarthria + dysphagia + preserved reflexes. Steroids + azathioprine long-term.

■ **MG on pyridostigmine: miosis + diarrhea + fasciculations + weak. Crisis type + drug action?**

Cholinergic crisis — HOLD pyridostigmine, atropine + airway per order.

MS: Autoimmune demyelination — relapses + remissions! Charcot triad (nystagmus + intention tremor + scanning speech) + Uhthoff (heat worsens). MRI plaques; steroids for relapse.

TRIAD + BLADDER + DRUGS

Feature	MS clue	Care
Vision	Optic neuritis — painful vision loss	Patch, steroids, eye rest
Motor	Spastic weak legs, foot drop	Baclofen + physio + AFO
Bladder	Spastic (urgency) or flaccid	Timed voids, self-cath
Heat	Hot bath worsens (Uhthoff)	Cool room, no hot tubs

- Relapse drugs: IV methylprednisolone; DMTs (interferon, glatiramer, natalizumab — PML watch); fatigue is #1 complaint — energy pacing.
- Swallow + speech therapy, pressure + fall bundle, depression + cognition screen, vaccines (avoid live in immunosuppressed).
- Lhermitte (neck flex → electric spine) + internuclear ophthalmoplegia are MS flags.

■ **FATAL TRAP**

Prescribing hot baths, overheated rooms or vigorous midday exercise to an MS patient precipitates pseudo-relapse with falls! Keep cool, schedule mornings, hydrate, report true relapse (>24h new deficit).

■ **NORCET HIT**

NORCET HIT: MS = young woman + relapsing diplopia/weakness + MRI Dawson fingers + oligoclonal bands. Steroids shorten relapse, DMTs cut frequency.

■ **Young woman with painful vision loss + leg spasticity + heat-worsening. Likely + confirmatory?**

MS — MRI brain/spine + LP oligoclonal bands + evoked potentials.

REHAB: No pulling the weak arm (subluxation)! Side-lying on AFFECTED side brief, then healthy side; swallow screen before ANY sip; neglect — teach from intact side.

POSITION + SWALLOW + NEGLECT

Problem	Action	Why
Shoulder sublux	Sling + pillow, NO overhead pull	Prevents tear/sublux
Foot drop	AFO + dorsiflexion exercises	Prevents contracture
Dysphagia	GUSS screen, chin-tuck, thickened	Stops aspiration pneumonia
Unilateral neglect	Plate/clock on intact side	Retrains scanning

- Turn q2h + air mattress; affected limbs in functional position (wrist extended, ankle 90°); constraint-induced therapy for learned non-use.
- Speech: short commands, pictures, no shouting; homonymous hemianopia — scan training + obstacle removal.
- Bladder: timed voids + prompted; bowel: fiber + fluids + routine; DVT: stockings + early mobilization.

■ **FATAL TRAP**

Lifting a hemiplegic patient by the weak arm or letting it dangle dislocates the shoulder and causes reflex sympathetic dystrophy! Support with sling/pillow, handle at scapula, ROM gently.

■ **NORCET HIT**

NORCET HIT: Right-brain stroke = left neglect + impulsivity; left-brain = aphasia + caution. Aspiration precaution = HOB 30–45° + suction ready.

■ **Left hemiplegia ignores left plate half + dangles weak arm. Two corrections?**

Support arm in sling/pillow (no pull) + place items on RIGHT (intact) + scan training.

IMMUNITY: CD4 <200 = AIDS + prophylaxis! Type I (anaphylaxis-IgE) is minutes; Type II cytotoxic, III immune-complex, IV delayed-cell. History + labs first.

LABS + 4 HYPERSENSITIVITIES

Test	Means	Action
CD4	<200 AIDS; <100 toxo/crypto risk	Prophylaxis + ART
Viral load	Treatment response	Adherence check
WBC/neutrophil	<500 severe neutropenia	Reverse isolation
IgE/eosinophil	Allergy/parasite	Trigger removal

- Gell-Coombs: I anaphylaxis (peanut/penicillin — epi NOW); II hemolytic (transfusion); III serum sickness/SLE; IV contact/TB test/T1DM.
- Assess: vaccine + allergy + infection + drug + travel + sexual history; lymph nodes, spleen, rashes, weight loss.
- Live vaccines banned in severe immunodeficiency; handwash + neutropenic diet + private room if ANC low.

■ FATAL TRAP

Giving live vaccine (BCG/MMR/varicella) or fresh flowers/raw salads to a severely neutropenic/AIDS patient seeds fatal infection! Check CD4/ANC, use killed vaccines, cooked food, filtered water.

■ NORCET HIT

NORCET HIT: I-Anaphylaxis/IgE-epi; II-cytotoxic-blood; III-complex-joints/kidney; IV-delayed-TB/contact. CD4 200/100/50 thresholds.

■ CD4 150 + thrush + weight loss. Stage + 2 actions?

AIDS (<200) — start/adjust ART + cotrimoxazole prophylaxis + OI screen.

■ **HIV spreads by blood + sex + mother-child — NEVER by hug/mosquito/sharing food! Needlestick PEP <72h (best <2h) x 28 days. Window period still infectious.**

TRANSMIT vs NOT + PEP

Spreads by	Does NOT spread by	Protect
Unprotected sex, needle share	Hugging, food, mosquito	Condoms + needle exchange
Mother-child (pregnancy/birth/milk)	Toilet, sweat, tears	ART in pregnancy + safe delivery
Blood transfusion (unscreened)	Cough, swimming pool	Screened blood + gloves

- PEP: wash (no squeezing), report, baseline HIV, start 3-drug <72h, 28 days, follow-up 6 wks/3 mo; side-effects + adherence counsel.
- Prevention: ABC (abstain/be faithful/condom) + PrEP for high-risk + male circumcision + STI treat + no needle reuse.
- Universal precautions ALWAYS: gloves + no recapping + sharps box + spill bleach 1:10.

■ **FATAL TRAP**

Squeezing a needlestick wound or delaying PEP beyond 72 hours lets virus establish!
Wash with soap-water, report, start PEP immediately — every hour counts, complete 28 days.

■ **NORCET HIT**

NORCET HIT: PEP <72h (ideal <2h). Window 2–12 wks antibody-negative but infectious.
ELISA screen → Western blot confirm (now NAT).

■ **Nurse needlestick from HIV+ source 1h ago. First 3 steps?**

Wash (no squeeze) + report + start PEP workup immediately (<2h best).

HIV: Acute flu → silent years → AIDS (CD4<200)! Thrush + PCP + TB + toxo + crypto kill. ART is lifelong combo — 95% adherence or resistance!

OI MAP BY CD4 + ART

CD4	OI flag	Prophylaxis/treat
<200	PCP (dry cough, ground-glass)	Cotrimoxazole
<100	Toxo (ring lesions), crypto	Cotrimoxazole + fluconazole watch
<50	MAC, CMV retinitis	Azithro + eye screen
Any	TB (India #1), thrush	ATT + nystatin/fluconazole

- ART: 2 NRTI + 1 NNRTI/INSTI (e.g., TLD); take same time daily; report rash/liver/jaundice/lactic signs; never stop abruptly.
- Monitor: CD4 + viral load + LFT/KFT + adherence diary + disclosure + partner testing + condoms even if undetectable (U=U counsel).
- Pregnancy: ART + elective C-section if load high + avoid breastfeeding per national guidance + nevirapine to baby per protocol.

■ **FATAL TRAP**

Stopping ART when feeling better or sharing doses breeds resistant virus and treatment failure! Adherence >95%, pill box + alarm + DOT, report vomiting/diarrhea that blocks absorption.

■ **NORCET HIT**

NORCET HIT: PCP = dry cough + exertional dyspnea + ground glass. Thrush scraping bleeds. Efavirenz = vivid dreams; zidovudine = anemia; tenofovir = kidney/bone.

■ **HIV CD4 120 + dry cough + dyspnea + ground-glass CT. Likely + drug?**

PCP pneumonia — high-dose cotrimoxazole + steroids + ART review.

NACO: Free HIV test + ART + PPTCT + condoms + needle exchange! 95-95-95 (diagnose-treat-suppress). ICTC tests, ART centres treat, blood banks screen.

SERVICE MAP

Centre	Does	Link
ICTC	Counselling + testing (opt-out ANC)	Confidential, consent
ART centre	Free ART + CD4/viral load	Adherence + LFU tracking
PPTCT	Mom + baby prophylaxis	Institutional delivery
Blood bank	ELISA + NAT screening	Voluntary donation

- Red Ribbon + targeted interventions (FSW/MSM/IDU/truckers/migrants); toll-free 1097 helpline.
- Stigma fight: confidentiality, no refusal of care, universal precautions (not isolation), partner notification with consent.
- Record: Nikshay for TB-HIV; LFU (lost to follow-up) retrieval is a nursing duty.

■ FATAL TRAP

Disclosing HIV status to family/employer without consent or refusing care drives patients underground and spreads infection! Confidentiality + non-judgmental counselling + linkage, never coercion.

■ NORCET HIT

NORCET HIT: NACO = National AIDS Control Org. 1097 helpline. PPTCT: ART to mother + syrup to baby + safe delivery + feeding counsel. Window donors deferred.

■ ANC mother HIV+ at ICTC. Programme + 2 linkages?

PPTCT + ART centre linkage + institutional delivery + infant prophylaxis.

■ **ANAPHYLAXIS: IM adrenaline 0.5 mg (0.5 mL 1:1000) mid-thigh NOW — no delay for IV line! Stridor + wheeze + hypotension after drug/food/sting = epi + O2 + fluids.**

ANAPHYLAXIS LADDER + AUTOIMMUNE FLAGS

Step	Action	Dose/point
1	IM adrenaline mid-thigh	0.3–0.5 mg 1:1000, repeat q5–15 min
2	O2 + lie flat legs up (unless vomit)	High-flow; no standing
3	IV fluids + antihistamine/steroid AFTER epi	Epi first, never instead
4	Observe 6–24h (biphasic)	Report + allergy band

- Autoimmune flags: SLE butterfly + photosensitivity; RA morning stiffness; myasthenia fatigability — steroids + immunosuppressants + infection watch.
- Allergy nursing: band + card + kit (epi-pen), desensitization education, asthma action plan, food-label reading.
- Transfusion allergy: STOP + saline + notify; keep epi + hydrocortisone at bedside for biologics.

■ **FATAL TRAP**

Giving IV antihistamine/steroid INSTEAD of IM adrenaline in anaphylaxis wastes the golden minutes and kills! Epi IM first, then fluids/O2/antihistamine; monitor for biphasic rebound hours later.

■ **NORCET HIT**

NORCET HIT: Epi 0.5 mg IM mid-thigh, repeat. Latex + penicillin + peanuts top triggers. SLE: ANA + sun avoid + kidney watch.

■ **Penicillin IV → wheeze + stridor + BP 80/40. First drug + route?**

IM adrenaline 0.5 mg mid-thigh + O2 + flat + call rapid response.

CANCER: CAUTION-UP warning + TNM staging + grade = plan! Biopsy is gold; pap/HPV/mammo screening finds curable early. Cachexia + paraneoplastic = red flags.

CAUTION + TNM + GRADE

Letter	Warning	Screen
C/A	Change bowel/bladder, A sore non-healing	Colonoscopy, oral exam
U/T	Unusual bleed, Thick lump	Pap/HPV, breast exam
I/O	Indigestion, Obvious wart/mole change	Endoscopy, dermoscopy
N/P	Nagging cough, Unexplained weight loss	Chest CT, weights

- TNM: T size/invasion, N nodes, M metastasis (M1 = stage IV); grade G1 well-diff (good) → G4 undiff (bad).
- Definitive = biopsy/histopathology; markers (PSA/CEA/CA-125/AFP) support, never alone; PET stages + watches response.
- Prevention: HPV + HBV vaccines, tobacco/alcohol stop, asbestos/UV avoid, obesity move; palliative from diagnosis.

■ **FATAL TRAP**

Reassuring a breast lump as 'just fibroadenoma' without triple assessment (exam + imaging + biopsy) misses curable cancer! Urgent referral <2 wks for red-flag lumps/bleeds/weight loss.

■ **NORCET HIT**

NORCET HIT: M1 = distant spread. Best prognosis = carcinoma-in-situ/screen-detected. Paraneoplastic: hypercalcemia + SIADH + Cushing + Trousseau.

■ **T2N1M0 breast cancer. What do letters mean + next gold test?**

T tumour size, N nodes+, M no distant spread — confirm with core biopsy + receptors.

HIGH-YIELD MAP: Breast + cervix + oral + lung + colorectal + prostate + stomach = India bulk! Node + bleed + weight loss + obstruction = workup.

SITE — FLAG — TEST

Cancer	Flag	First test
Breast	Painless lump + dimpling + nodes	Mammo + US + core biopsy (ER/PR/HER2)
Cervix	Post-coital bleed	Pap + VIA + HPV + colposcopy
Oral	Non-healing ulcer + tobacco	Biopsy + neck exam
Lung	Cough + hemoptysis + clubbing	CT + bronchoscopy biopsy
Colorectal	Alternating bowel + blood	Colonoscopy + CEA
Prostate	Nocturia + retention + back pain	PSA + DRE + TRUS biopsy

- Stomach: Virchow (left supraclavicular) + Sister Mary Joseph nodules; liver: AFP + HBV; ovary: CA-125 + bloating.
- Childhood: leukemia (pallor + bleed), retinoblastoma (white reflex), Wilms (flank mass) — early referral cures.
- Nursing: screening camps, tobacco cessation, HPV vaccine 9–14 yrs, stoma + wound + pain + nutrition links.

■ **FATAL TRAP**

Doing PR exam vigorously or massaging a suspected pheochromocytoma/testicular mass can precipitate crisis/spread! Gentle exam, vitals first, urgent oncology referral — never squeeze.

■ **NORCET HIT**

NORCET HIT: Painless + progressive + weight loss = malignancy until proven. Left supraclavicular = abdominal primary. Postmenopausal bleed = endometrial cancer.

■ **Post-coital bleeding + foul discharge. Likely + screen trio?**

Cervical cancer — Pap + VIA + HPV-DNA + colposcopy/biopsy.

CHEMO: Nadir Day 7–14 (infection/bleed peak)! Vesicant leak = STOP + aspirate + antidote. Triple antiemetic (5HT3 + NK1 + dexta) for highly emetogenic.

VESICANT + NADIR + EMESIS

Risk	Rule	Antidote/care
Vesicant (vincristine/doxo)	Free flow + check blood return	STOP, aspirate, elevate, plastic surgery
Nadir neutropenia	ANC <500 = danger	Neupogen, reverse isolation, no live vaccine
Thrombocytopenia	<20k spontaneous bleed	No IM/razor/floss-harsh
Emesis (cisplatin)	Triple prophylaxis BEFORE	Ondansetron + aprepitant + dexta

- Handle: gloves/gown/goggles, closed system, spill kit, double-check BSA + allergy + consent + pregnancy test.
- Specifics: vincristine IV ONLY (intrathecal fatal); bleomycin → lung fibrosis (bleomycin flag); doxorubicin → cardiac (ECHO + lifetime cap).
- Mucositis: soft brush + saline rinses, ice chips (cryotherapy), nystatin if thrush; alopecia wig + scalp cooling counsel.

■ **FATAL TRAP**

Flushing a vesicant extravasation or applying heat to vincristine leak spreads necrosis! STOP infusion, aspirate, antidote (hyaluronidase/dexrazoxane per protocol), cold (except vincristine-warm), elevate, mark + photo + surgeon.

■ **NORCET HIT**

NORCET HIT: Nadir 7–14 days. Febrile neutropenia = cultures + broad antibiotics <1h. Hemorrhagic cystitis (cyclophosphamide) → MESNA + fluids.

■ **Day-10 chemo + fever 38.5 + ANC 300. Diagnosis + first hour?**

Febrile neutropenia — cultures + broad-spectrum antibiotics within 1 hour.

RADIO: Skin erythema → dry → moist desquamation! Wash gently, NO metal/tape/perfume. Sealed implant = TIME-DISTANCE-SHIELD + private room + dosimeter.

EXTERNAL vs BRACHYTHERAPY

Type	Care	Safety
External beam	Gentle wash, pat dry, aloe per order	No shaving cream/metal, loose cotton
Brachy sealed	Bed rest, low-residue, Foley	Private room, limit time, behind shield
Unsealed I-131	Isolation + no pregnant visitors	Flush twice, separate laundry
Mucositis/diarrhea	Saline rinses, BRAT + fluids	Report bloody stool

- Markings: do NOT wash off ink lines; no deodorant/powder on field; sunscreen lifelong on field.
- Fatigue peaks 2 wks post-course — pace energy; fertility + pregnancy counsel before start; blood counts weekly.
- Brachy dislodge (vaginal applicator out) = DO NOT touch with hands — long forceps + lead pot + call RSO + evacuate room.

■ FATAL TRAP

Picking at moist desquamation or applying adhesive tape on irradiated skin tears it into infected ulcer! Pat dry, air-expose, hydrogel per order, report blister/pus promptly.

■ NORCET HIT

NORCET HIT: ALARA — time short, distance 2m, shield lead. Dislodged source = forceps + lead + RSO. Diarrhea + cystitis are pelvic-RT flags.

■ Cervix brachytherapy applicator found in bed. First action?

Do NOT touch by hand — long forceps to lead container, clear room, call RSO/physicist.

SURGERY cures early + debulks late! BMT nadir needs sterile + GVHD watch (skin-gut-liver). Immunotherapy colitis/pneumonitis = hold + steroids.

MODALITY — NURSING

Modality	Nursing must	Watch
Curative resection	Consent + stoma marking + DVT bundle	Margins + nodes + drain
Debulk/palliative	Pain + obstruction relief goals	Family + advance care
BMT/HSCT	Sterile + HEPA + no plants	GVHD rash/diarrhea/jaundice
CAR-T/checkpoint	Fever + neuro check (ICANS)	CRS hypotension — tocilizumab

- BMT phases: conditioning (chemo/TBI) → nadir (ANC zero) → engraftment (WBC rise) — reverse isolation + cooked food + no sick contacts.
- GVHD: skin rash + watery diarrhea + cholestasis → tacrolimus/cyclosporine + levels + kidney/BP watch.
- Lymphedema post-mastectomy/axillary: no BP/needle on arm, elevate, sleeve, physio; gene therapy consent + long follow-up.

■ **FATAL TRAP**

Taking BP or IV on the mastectomy/lymphadenectomy arm precipitates lifelong lymphedema! Tag arm, use opposite side, elevate + sleeve, report heaviness/tight ring early.

■ **NORCET HIT**

NORCET HIT: BMT match = HLA. Engraftment ~2–4 wks. CRS (fever + hypotension post CAR-T) = tocilizumab + ICU. Neutropenic diet cooked only.

■ **Day-20 BMT + macular rash + watery diarrhea + bilirubin up. Suspect + drug class?**

Acute GVHD — calcineurin inhibitor/steroids + biopsy + levels.

■ **EMERGENCIES: SVC (face swell + sit up) | Cord compression (back pain + saddle = MRI <24h) | TLS (K/Phos up, Ca down) | Sepsis — all STAT!**

4 KILLERS + FIRST MOVE

Emergency	Flag	First move
SVC syndrome	Face/neck swell + stridor	Sit up + O2 + diuretics + stent/RT
Cord compression	Back pain + weak + retention	Flat + steroids + MRI <24h + decompress
Tumour lysis (TLS)	K/Phos/Uric UP, Ca DOWN + AKI	fluids + allopurinol/rasburicase + labs q6h
Febrile neutropenia	Fever + ANC <500	Cultures + antibiotics <1h

- Hypercalcemia: fluids + furosemide + bisphosphonate + mobilize; SIADH: fluid restrict + hypertonic if seizing.
- DIC: bleed + clot together — platelets + FFP + heparin per order; anaphylaxis kit for biologics.
- Document neuro + I/O hourly; keep rasburicase + calcium + insulin-glucose ready in TLS high-risk (bulky leukemia/lymphoma).

■ **FATAL TRAP**

Laying flat an SVC-obstructed patient or delaying MRI for cord compression paralyzes!
SVC — sit upright + O2; cord — steroids + emergent MRI/decompression within 24h or permanent paraplegia.

■ **NORCET HIT**

NORCET HIT: TLS labs — K UP, Phos UP, Uric UP, Ca DOWN. Cairas: Cage (SVC sit), Cord (steroid+MRI), Lysis (flush+allopurinol), Sepsis (1h antibiotics).

■ **Lymphoma chemo Day-2: K 6.1 + uric high + Ca low + oliguria. Emergency + fluids/drug?**

Tumour lysis — aggressive fluids + allopurinol/rasburicase + q6h labs + nephrology.

PALLIATIVE starts at diagnosis, HOSPICE last 6 months! Pain ladder + bowel + breath + dignity + family + grief. Total pain = physical + psycho + social + spiritual.

LADDER + DYSPNEA + GRIEF

Step	Drug	Nursing
1 mild	Paracetamol/NSAID	Liver/kidney watch
2 moderate	Tramadol/codeine + antiemetic	Constipation bundle
3 severe	Morphine/fentanyl patch	Rescue dose + naloxone ready
Adjuvant	Amitriptyline/gabapentin/dexa	Neuropathic/bone pain

- Constipation bundle with EVERY opioid: fluids + fiber + senna + docusate + walk; laxative before enema.
- Dyspnea: fan + upright + O2 + low-dose morphine + anxiolytic; death rattle — glycopyrrolate + reposition + family counsel.
- Grief: Kubler-Ross (denial–anger–bargain–depress–accept); dignity: wishes + rituals + will + legacy; caregiver respite + bereavement follow-up.

■ FATAL TRAP

Withholding morphine fearing addiction in terminal cancer leaves needless suffering and hastens decline! Titrate to pain + sedation scale, bowel + nausea covered, addiction rare in true pain — comfort is the goal.

■ NORCET HIT

NORCET HIT: Breakthrough = 1/6th daily dose. Fentanyl patch q72h (no heat!). Hospice <6 mo prognosis; palliative anytime. DNR needs informed consent + docs.

■ Terminal cancer on morphine, no stool 4 days + colicky pain. Cause + bundle?

Opioid constipation — fluids + fiber + senna + docusate + mobilize + manual check.

■ **EMERGENCY: ABCDE every time — Airway, Breathing, Circulation, Disability, Exposure! Golden hour + polytrauma team + hemorrhage control decide survival.**

ABCDE + HEMORRHAGE

Step	Assess	Act
A	Stridor, snoring, foreign body	Jaw thrust, suction, adjunct, intubate
B	RR, SpO ₂ , chest rise, pneumothorax	O ₂ , seal sucking wound, needle decompress
C	Pulse, BP, bleed, pelvis	Direct pressure, TXA <3h, 2 large IVs, FAST
D	GCS, pupils, glucose, lateralize	C-spine, CT head if low
E	Temp, rashes, log-roll, wounds	Warm, tetanus, antibiotics

- Polytrauma: team huddle + roles, massive transfusion 1:1:1, permissive hypotension (except TBI), pelvic binder if unstable.
- AMPLE history + tetanus + pain + NPO + Foley/NG after clearance; family desk + valuables log.
- Reassess after EVERY intervention — deterioration = missed bleed/tension/pump failure.

■ **FATAL TRAP**

Log-rolling a suspected spine injury with one nurse or flexing neck to 'check stiffness' converts fracture to permanent cord cut! 3–4 staff log-roll, rigid collar till cleared, backboard time minimal.

■ **NORCET HIT**

NORCET HIT: TXA within 3h of bleeding trauma. C-A-B if arrest (compress first). Pelvic fracture + hypotension = binder + blood. Mnemonic ABCDE never skip.

■ **Bike crash + open femur bleed + RR 32 + BP 88/60. First 3 moves?**

Direct pressure + O₂ + 2 large-bore IVs + TXA + trauma team + FAST.

SNAKEBITE: Keep limb IMMOBILE at heart level + NO tourniquet/cut/suck! 20-min whole-blood clot test + ASV per grade. Dog bite = wash 15 min + ARV + RIG.

SNAKE vs SCORPION vs DOG

Bite	Flag	Nursing
Neurotoxic (krait/cobra)	Ptosis, dysphagia, resp weak	ASV + neostigmine + vent ready
Vasculotoxic (viper)	Bleed, blister, AKI	ASV + clot test + dialysis watch
Scorpion	Sweating + tachy + priapism/pain	Prazosin + ice + ICU
Dog (rabies)	Licks/breaks skin	Wash 15 min + ARV days 0-3-7-14-28 + RIG

- Snake: reassure + immobilize + remove rings, mark swelling edge with time, no NSAIDs/IM (bleed), ASV test dose watch + reaction kit.
- 20WBCT: blood in glass, no clot at 20 min = coagulopathy — repeat q6h; urine for red/Brown + K+ (hemolysis).
- Rabies: category II (nibble, vaccine) vs III (transdermal, vaccine + RIG into wound); counsel stray-dog + pre-exposure for vets.

■ **FATAL TRAP**

Tying tight tourniquet, cutting/sucking or applying herbal paste on snakebite concentrates venom and rots tissue! Pressure-immobilize (neurotoxic), rapid transport, ASV only in facility with reaction cover.

■ **NORCET HIT**

NORCET HIT: Krait night painless + morning paralysis. Viper = bleed + AKI. Scorpion prazosin first. Rabies wash 15 min soap-water + povidone.

■ **Viper bite + oozing gums + 20WBCT unclotted. Meaning + drug?**

Coagulopathy — ASV per grade + clotting/DIC/AKI monitoring + blood bank.

POISON: Airway + charcoal <1h (if protected) + antidote! **OPC = SLUDGE + atropine + pralidoxime; paracetamol = NAC; never induce vomiting for corrosives!**

ANTIDOTE TABLE (EXAM GOLD)

Poison	Flag	Antidote
OPC/carbamate	SLUDGE + miosis + fasciculation	Atropine + pralidoxime (OPC)
Paracetamol	Silent 24h then liver fail	N-acetylcysteine
Iron	Vomit + X-ray tablets	Desferrioxamine
Methanol	Blind + acidosis	Ethanol/fomepizole + dialysis
Corrosive acid/alkali	Burns lips, drool	NO vomit/lavage — dilute + scope

- Decontaminate: remove clothes + wash skin, gastric lavage <1h if airway safe, charcoal if <1h + bowel sounds + no hydrocarbon/corrosive.
- Enhance elimination: fluids + urine alkali (salicylates), dialysis (methanol, lithium); observe 6–24h (delayed).
- Psych: 1:1 watch, stomach contents saved/labelled, police/MLC documentation, suicide-risk counselling post-recovery.

■ **FATAL TRAP**

Inducing vomiting or charcoal in kerosene/corrosive ingestion aspirates/burns twice and kills! Keep NPO upright, O2, small sips only per order, urgent endoscopy/surgery — lavage only if protected airway + indicated.

■ **NORCET HIT**

NORCET HIT: OPC smell garlic + pinpoint + wet lungs. Atropine till dry (tachy ok). Paracetamol Rumack nomogram + NAC <8h best. Keep sample + wrapper.

■ **OPC farmer + pinpoint + bronchorrhea + fasciculations. Two drugs?**

Atropine (dry lungs) + pralidoxime (reactivate) + airway + decontamination.

HEATSTROKE (>40C + dry + confused) = ICE + fluids STAT! Hypothermia (<35C) = gentle + slow rewarm — rough handling triggers VF!

HEAT vs COLD

Feature	Heatstroke	Hypothermia
Temp/skin	41C, HOT-DRY-red	<35C, cold-pale-stiff
Mental	Confused/seizing	Drowsy then arrhythmia
Cool/rewarm	Ice bath + fans + fluids	Remove wet, blanket, warm IV 38C
Drugs	Dantrolene if malignant	No IM/sedatives if cold-slow

- Heat: stop activity + shade + strip, ice neck/groin/axilla, O2 + LR, bladder temp, seizure + DIC + AKI watch; sports/march counselling.
- Cold: handle gently (VF risk), no alcohol massage, warm humid O2, active rewarm only in ICU (arrest needs prolonged CPR + ECMO).
- Frostbite: rewarm 37–39C water, no rubbing, blisters intact, tetanus + aloe + ibuprofen per order.

■ FATAL TRAP

Giving antipyretic (paracetamol) for heatstroke or rubbing frostbitten hands with snow worsens injury! Heatstroke needs physical cooling + fluids; frostbite needs controlled warm bath + no friction.

■ NORCET HIT

NORCET HIT: Exertional (young, sweaty) vs classic (elderly, dry). Hypothermia Osborn J-wave + bradycardia. Rewarm shock needs ICU.

■ Marathon + 41.5C + dry + GCS 10. Diagnosis + first cool?

Heatstroke — ice-water immersion + evaporative cooling + fluids + ABCs.

DISASTER: Mitigate-prepare-respond-recover! Hospital incident command + decontamination + surge beds. Nurse = triage + tags + tracking + family desk.

CYCLE + COMMAND

Phase	Nursing role	Kit
Mitigation	Hazard map, drills	Fire exits, quake straps
Preparedness	Rosters, stock, mock drills	Crash carts, O2 manifold
Response	Triage + tags + decontam	PPE, stretchers, blood
Recovery	Rehab + records + counsel	PTSD screen, claims

- NDMA (national) → SDMA → DDMA chain; hospital HICS: commander + medical-tech + logistics + finance.
- Decontamination: remove clothes (80% off), wash head-to-toe, contain runoff; staff PPE + rotation + psych breaks.
- Documentation: tag + register + photo + MLC + dead-body labelling + handover chain intact.

■ **FATAL TRAP**

Rushing into chemical/radiation scene without PPE and eating/drinking in hot zone poisons responders! Stage upwind, full PPE, buddy system, decon before hospital entry.

■ **NORCET HIT**

NORCET HIT: NDMA apex. Mock drills quarterly. Crush syndrome (earthquake) — fluids + K+ + compartment watch. PTSD debrief <72h.

■ **Chemical leak + victims arriving unwashed at ER gate. First nursing order?**

Stop entry — outside decon corridor + PPE + triage before admission.

TRIAGE: Sort to save MOST! START (adult): RPM <30-2-can do. Red immediate, Yellow delayed, Green walking, Black expectant/dead. Retriage often!

START + TAGS

Tag	Means	Example
RED	Immediate, survivable critical	Airway, bleed, tension, shock
YELLOW	Delayed, stable for now	Closed fracture, controlled bleed
GREEN	Walking wounded	Minor cuts, sprains
BLACK/GREY	Expectant / dead	Agonal, unsurvivable, pulseless

- START: Respirations (no after open = black; >30 = red), Perfusion (cap refill >2s/no radial = red), Mental (cannot follow = red).
- JumpSTART (child): 5 rescue breaths if apneic + pulse; infants need Broselow tape + weight dosing.
- Nurse: tag visible + time + name/number, reassess q15min, family desk separate, dead-body area screened + documented.

■ FATAL TRAP

Doing CPR on pulseless-apneic in mass-casualty START (black) while reds bleed out wastes the golden hour! Follow RPM, tag honestly, treat reds first — retriage as resources change.

■ NORCET HIT

NORCET HIT: RPM = Respirations-Perfusion-Mental. Walking = green. Retriage is mandatory. SALT alternative: Sort-Assess-Lifesave-Treat.

■ Blast: apneic after airway open + no pulse. Tag + next?

Black (expectant/dead) — move on to salvageable reds.

MLC: Every injury/poison/burn/assault/bite/death-unknown = MLC! Inform police, preserve clothes/bullets/swabs, chain intact, never guess time/cause.

MLC DO + DON'T

Do	Don't	Why
Inform MO + police + register	Declare dead on arrival cause	Court + autopsy
Label + seal + sign samples	Wash clothes/bullets	Evidence lost
Body diagram + photos + vitals	Alter wounds	Tamper charge
Handover receipt + counsel	Hand body without papers	Liability

- Dying declaration: magistrate preferred; nurse witness + exact words + consciousness note if emergency.
- Consent: informed + witness for procedures; assault — explain exam + offer chaperone + STD/pregnancy prophylaxis.
- Records: date-time-signature-stamp, overwrites countersigned, MLC file locked + copies per protocol.

■ **FATAL TRAP**

Washing blood-stained clothes or throwing away bullets/swabs destroys evidence and invites prosecution! Bag-paper (not plastic for wet), seal-label-sign, refrigerate swabs, chain register unbroken.

■ **NORCET HIT**

NORCET HIT: Brought-dead still MLC + police + autopsy. Poison sample + wrapper saved. Assault kit + consent + chaperone.

■ **Assault victim + torn clothes + police not yet told. First 3?**

Treat ABCs + preserve clothes/samples + inform MO/police + document.

AGING: Normal loss, NOT disease! CGA (medical + functional + psycho + social + env) + Katz/ADL + falls + polypharmacy review = independent longer.

SYSTEM AGING + CGA

System	Normal aging	Nursing tweak
Heart/lung	Stiff, slow, low reserve	Pace, sit, O2 wean slow
Kidney	GFR falls, thirst blunted	Fluids + drug-dose cut
Brain/senses	Slow, presbyopia/cusis	Loud-clear, light, hearing aid
Skin/bone	Thin, osteo, falls	Hip protectors, Vit D + rails
Bladder/bowel	Urgency, constipation	Timed voids, fiber

- CGA domains: comorbidity + meds + ADL/IADL (Katz/Lawton) + gait (Timed Up-Go) + cognition (MMSE) + mood (GDS) + nutrition (MNA) + home safety.
- Polypharmacy: Beers criteria (avoid sedatives/anticholinergics), brown-bag review, one pill box + large labels.
- Programmes: NPHCE + Ayushman + pension + rail concession + helpline 14567; home visits + day-care linkage.

■ FATAL TRAP

Calling delirium 'just old age' or sedating wandering with restraints/antipsychotics hides UTI/pain/drugs and kills! CAM screen + hydrate + toilet + glasses/aids + day-night rhythm first.

■ NORCET HIT

NORCET HIT: CGA gold. Falls screen Morse/Hendrich. Presbyopia (lens) + presbycusis (high-tone loss). Thirst down — push fluids.

■ 78-yr + new confusion + UTI + on sedative. Delirium vs dementia + move?

Acute delirium — treat UTI + review drugs + hydrate + reorient, no restraint.

ELDER CARE: Dignity + consent + privacy at EVERY age! Abuse (bruise-burn-fear) = separate + examine + report. Caregiver burnout = respite or collapse.

ABUSE FLAGS + CAREGIVER

Abuse	Flag	Act
Physical	Grip bruises, burns, fear	Separate, photo, MLC, shelter
Neglect	Bedsore, dehydration	Admit + nutrition + plan
Financial	Missing pension, forced sign	Bank + legal aid
Verbal	Crying, withdrawn	Counsel + family meet

- Sexuality: normal need — privacy + safe-sex + prostate/menopause counsel; never mock; STI screen if risk.
- Stress: loss + loneliness — day-care + bhajan/prayer + pets + sleep hygiene; depression (GDS) vs dementia screen.
- Caregiver: Zarit burden + respite + aids (walker, commode, diapers) + home-nurse + DBT/bathing training; aids-prosthesis fit + pressure care.

■ **FATAL TRAP**

Interviewing suspected abuse victim in front of attendant or promising absolute secrecy traps them longer! Separate privately, believe, document exact words + body map, report per law + shelter.

■ **NORCET HIT**

NORCET HIT: MWPSA Act 2007 — children must maintain parents; helpline 14567. Restraints need order + least-time + checks.

■ **Elder + grip bruises + flinches at son + says fell. Next?**

Separate privately + head-toe exam + MLC + social work + safe discharge plan.

HOME CARE: Bed-chair-toilet same floor + rails + light + bell! Aids fitted + skin + physio + vaccines (flu/pneumo) + Ayushman card = age in place.

HOME KIT + PROGRAMMES

Need	Provide	Teach
Mobility	Walker, stick, wheelchair	Brakes + transfers
Toileting	Commode, rails, diapers	Skin + UTI watch
Senses	Specs, hearing aid, dentures	Clean + spare battery
Safety	Bell, light, non-slip, fire alarm	Fall + burn drill

- Visits: vitals + meds + wound + catheter + physio + labs; red-flag card (chest pain/stroke/fall/fever) + ambulance number.
- Vaccines: annual flu + pneumococcal + COVID + shingles per age; nutrition: protein + calcium + Vit D + denture-soft diet.
- Link: NPHCE geriatric clinics + Ayushman + IGNOAPS pension + rail/air concession + senior associations.

■ FATAL TRAP

Leaving elder alone on high bed without rails/bell or with loose rugs/cords guarantees hip fracture! Low bed + both rails + call bell + night light + non-slip + supervised bath.

■ NORCET HIT

NORCET HIT: Home safety bundle = light + rails + bell + non-slip + fire. Denture overnight OUT + cup. Hearing aid whistle = refit.

■ Discharge elder alone upstairs + walker unfitted + no bell. Fix trio?

Same-floor bed + fit walker + rails/bell/light + home-nurse + helpline card.

ICU: ABCDE + monitors + bundles! EWS/MEWS + q1h vitals + I/O + GCS + pain + delirium (CAM-ICU) + skin + lines = catch crash early.

MONITOR + BUNDLES

Monitor	Target/alarm	Act
SpO2/ABG	94–98%, PaO2 >80	O2 titrate, suction prn
MAP/CVP	MAP ≥65, CVP 8–12	Fluids/pressors
GCS/pupils	Trend + aniso	CT + neuro call
I/O + weight	1 mL/kg/hr urine	AKI bundle + diuretics

- ABCDEF bundle: Airway, Breathing trial, Choice sedation, Delirium, Early mobility, Family; VAP + CLABSI + CAUTI bundles daily.
- Staffing: 1:1 ventilated, 1:2 HDU; handover SBAR + read-back + double-check highs (insulin/heparin).
- Family: open hours + diary + goals + breaking-bad-news (SPIKES) + end-of-life wishes.

■ FATAL TRAP

Silencing alarms or charting vitals without seeing patient misses tamponade/sepsis and codes! Respond <60 sec, trend EWS, escalate + document — never mute-and-walk.

■ NORCET HIT

NORCET HIT: MEWS ≥5 = rapid response. Daily SAT + SBT cuts vent days. CAM-ICU delirium + RASS sedation target -2 to 0.

■ Post-op MEWS 6 + cold + urine 15 mL/hr. Meaning + team?

Deteriorating (shock) — rapid response + ABCs + fluids/cultures + HDU/ICU.

VENT: AC (all breaths) vs SIMV (some) vs CPAP/PS (spontaneous)! Low tidal 6 mL/kg + PEEP + FiO2 wean + DVT/stress prophylaxis + daily SBT.

MODE + SETTING + WEAN

Mode	When	Watch
AC/CMV	No effort, arrest	Hypervent + fight — sedate
SIMV + PS	Weaning bridge	Sync + work of breathing
CPAP/PSV	Spontaneous, trial	Fatigue — back to AC
PEEP	ARDS, low PaO2	BP drop + barotrauma

- Settings: Vt 6–8 mL/kg IBW, RR 12–20, FiO2 lowest for SpO2 94–98, PEEP 5 start; ABG + plateau <30 cmH2O.
- Care: HOB 30–45°, suction only when needed (<10s + pre-O2), cuff 20–30, mouth + eye + DVT + ulcer bundle.
- Wean: daily SAT + SBT (T-piece/CPAP 30–120 min) + RSBI <105 + cough + secretions manageable → extubate + stridor watch.

■ **FATAL TRAP**

Suctioning routinely q15min or bagging with 100% O2 long desiccates + infects + crashes pressure! Suction on indication, sterile closed system, pre-oxygenate briefly, hyperinflate only per order.

■ **NORCET HIT**

NORCET HIT: Unplanned extubation = retape + midline + bite-block + restraints-last + wean fast. Cuff leak + stridor = racemic epi + steroids.

■ **ARDS Vt 10 mL/kg + plateau 34. Fix?**

Lower Vt to 6 mL/kg IBW + PEEP per table + plateau <30 + prone if severe.

ALARM? DOPE it — Displacement, Obstruction, Pneumothorax, Equipment! High = block/kink/fight; Low = leak/disconnect. Bag + call, never mute!

HIGH vs LOW + DOPE

Alarm	Likely	Bedside fix
High pressure	Kink, mucus, bite, bronchospasm	Unkink, suction, bronchodilator
Low pressure/volume	Leak, disconnect, cuff	Reconnect, reinflate cuff
Apnea	Sedation, disconnect	Bag + reconnect + rate
FiO2/power	O2 fail, circuit wet	Backup O2 + drain water

- DOPE walk: tube Depth (21–23 cm teeth) + Obstruct (suction) + Pneumo (absent breath + trachea shift) + Equipment (circuit/power).
- If collapse: disconnect + bag 100% + call anaesthesia + prepare reintubation + chest drain tray if tension.
- Document: alarm + finding + action + vitals/ABG after; keep suction + ambu + spare tube at head always.

■ **FATAL TRAP**

Reconnecting a contaminated circuit or pushing tube deeper blindly during low-pressure alarm aspirates + mainstems! Bag with sterile + check depth/cuff/circuit, X-ray confirm, sterile suction only.

■ **NORCET HIT**

NORCET HIT: Tube 21–23 cm adult. Cuff 20–30 cmH₂O. High = obstruction; Low = leak. Tension = needle 2nd ICS + drain.

■ **High-pressure alarm + biting + SpO2 fall. First 2?**

Bite-block + unkink + suction per need + sedate/sync per order + ABG.

SHOCK: Fluids if CVP low + warm; pressors if MAP <65! Defib UNSYNC for VF/pulseless VT (biphasic 120–200J); cardiovert SYNC for unstable tachy WITH pulse.

PRESSOR + SHOCK CURRENT

Drug	Use	Watch
Noradrenaline	Shock first pressor	Extravasation — large vein
Adrenaline	Arrest + anaphylaxis	Tachy + lactate
Dopamine/dobutamine	Brady + pump fail	Arrhythmia
Amiodarone	VF/VT refractory	Thyroid + lung + QT

- Defib: pads (no gel gaps), clear + O2 off, shock + immediate CPR 2 min; pacing: capture + output + sedation + check pulse.
- Cardioversion SYNC 50–100J (AF/SVT unstable) + sedation + anticoag check; TCP bridge to transvenous.
- Post-ROSC: TTM + MAP + glucose + seizure watch + debrief + family update.

■ **FATAL TRAP**

Defibrillating asystole/PEA or cardioverting without SYNC (VF protocol on perfusing tachy) burns and fails! Shock only VF/pulseless VT unsync; unstable WITH pulse sync + sedate; asystole = CPR + epi + cause hunt.

■ **NORCET HIT**

NORCET HIT: Biphasic 120–200J, monophasic 360J. Pads 10 cm apart. Transcutaneous pacing + atropine for symptomatic brady.

■ **Monitor VF + pulseless + O2 on. Shock steps?**

Clear + O2 off + unsync 120–200J biphasic + immediate CPR 2 min + epi/amio.

HIGH-ALERT: Double-check heparin/insulin/norad + label + pump lock! Central line = sterile + daily need + remove ASAP. Crash cart checked every shift.

LINE + PUMP + CART

Line	Site/care	Remove when
Central	Subclavian/jugular, full sterile	Fever/CLABSI or no need
Arterial	Radial, Allen test, zero level	Bleed/spasm/infection
Infusion pump	Drug + dose + rate + lock	Bolus only per order
Crash cart	Defib + airway + drugs in date	After every use/shift

- CLABSI bundle: hand + chlorhexidine + full drape + avoid femoral + daily review; scrub hub 15s + flush.
- Pump: independent double-check, allergy + weight + limits programmed, handover rate + remaining + line patent.
- Resus equipment: suction + ambu + O2 + laryngoscope light + ET sizes + stylet + capnography ready.

■ **FATAL TRAP**

Flushing a blocked central line forcefully or running norad peripherally long embolizes/clots and necroses skin! Aspirate + reposition + urokinase per order; extravasation — stop + phentolamine + elevate + surgeon.

■ **NORCET HIT**

NORCET HIT: Zero arterial at phlebostatic axis (4th ICS mid-axillary). Damping = kink/clot. Trolley: airway-breathing-circulation-drugs-docs.

■ **Norad extravasation + white-cold patch. Action?**

Stop + aspirate + elevate + phentolamine per order + new large/central access.

■ **SEPSIS: qSOFA + lactate + cultures + broad antibiotics <1h + 30 mL/kg fluids + norad for MAP! Shockable (VF/VT) = shock; non-shockable = epi + causes.**

SEPSIS-3 + ACLS CORE

Shock	Clue	Fluid/pressor
Septic	Fever + low + lactate high	30 mL/kg + norad
Cardiogenic	Wet lungs + cold + JVP up	Dobut + diuretics, no flood
Hypovolemic	Dry + bleed/trauma	Blood + TXA + cause
Anaphylactic	Wheeze + swell + low	IM epi + fluids

- Sepsis bundle: lactate + 2 cultures BEFORE antibiotics + broad <1h + fluids + pressors + source (drain/abx) + steroids if refractory.
- ACLS: push hard-fast (5–6 cm, 100–120) + minimize breaks + shock VF/VT + epi 1 mg q3–5 min + amio + Hs/Ts hunt.
- Post: MAP + urine + lactate clearance + glucose + VTE/ulcer prophylaxis + rehab + sepsis follow-up.

■ **FATAL TRAP**

Pouring litres into cardiogenic shock or delaying antibiotics for CT in septic shock drowns/fails! Bedside echo + lungs + JVP guide fluids; cultures then antibiotics <1h — source control same day.

■ **NORCET HIT**

NORCET HIT: qSOFA (RR≥22 + GCS<15 + SBP≤100). Lactate >2 = shock. Hs/Ts (hypoxia, hypoK, tamponade, tension, toxins).

■ **Fever + RR 28 + SBP 88 + lactate 4. Bundle first hour?**

Cultures + broad antibiotics + 30 mL/kg + lactate + norad + source hunt.

DAILY: Wake + breathe + move + feed + clean lines! Delirium-free + pressure-free + clot-free + home-ready. SBAR + meds + pending = safe handoff.

DAY + HANDOVER

Daily	Do	Tick
Wake	SAT + RASS + delirium	CAM-ICU
Breathe	SBT + chest physio	RSBI
Feed/move	Early protein + dangle/walk	GCS + drains
Lines	Need? remove today	CLABSI/CAUTI/VAP

- Skin: Braden + turns + heel off + barrier; eyes: tape + tears if coma; mouth: chlorhexidine + brush.
- Transition: step-down criteria + summary + meds reconciled + follow-up + warning card + family demo (suction/feed/dressing).
- Handover: SBAR + vitals trend + infusions + pending labs/consults + code status + family updated.

■ **FATAL TRAP**

Leaving Foley/CVC 'just in case' after ICU need breeds CAUTI/CLABSI and resists transfer! Remove lines/catheters ASAP, void trial + output watch, transfer with summary + drugs + contacts.

■ **NORCET HIT**

NORCET HIT: ABCDEF liberation. ICU psychosis = delirium (day-night + glasses/aids + sleep). Family conference before shift to ward.

■ **Ward transfer + Foley Day-6 + no order + afebrile. Action?**

Review need — remove + void trial + perineal care + document output.

DIGNITY: Truth + consent + DNR documented + comfort + rituals! Workers: hazard map + PPE + surveillance + TB/silicosis/lead watch + ESI care.

EOL + WORKER MAP

Issue	Rule	Nursing
DNR/advance directive	Written + witness + team	Comfort + family + chaplain
Breaking bad news	SPIKES private + empathy	Stay + listen + plan
Pneumoconiosis	Dust + X-ray eggshell	Mask + spirometry + TB screen
Lead poisoning	Burton line + colic + wrist drop	Chelate + source remove + ESI

- EOL: pain + breath + secretions + mouth + skin + spiritual + will + organ-donation ask sensitively.
- Worker: pre-placement + periodic exam, audiometry (noise), lead/blood, pesticides cholinesterase, needle + radiation badges.
- ESI + factories Acts: compensation + rest sheds + crèche + first-aid room + nurse role in health education.

■ **FATAL TRAP**

Continuing futile CPR against documented DNR or hiding diagnosis from dying patient breaches autonomy and law! Confirm code status each shift, honour directives, comfort + dignity + family presence.

■ **NORCET HIT**

NORCET HIT: SPIKES (Setting-Perception-Invitation-Knowledge-Empathy-Summarize). Silicosis-TB link. Byssinosis Monday chest (cotton). ESI dispensary care.

■ **DNR + Cheyne-Stokes + rattle + family panics. Comfort trio?**

Reposition + suction prn + low O2/morphine per order + stay + explain + rituals.